模式识别课程

Thinking with Images
以图思考



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课程提纲

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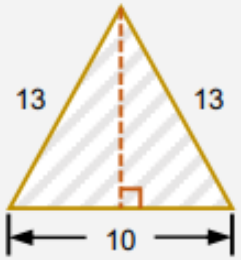
一、课题背景介绍
二、工具驱动的视觉探索
三、程序化视觉操作
四、原生视觉想象
五、总结与展望

背景介绍

借鉴 LLM 的思维链 (CoT) 范式，多模态大模型 (LMMs) 视觉推理能力取得进步

Question:

Find the area of the isosceles triangle in the image.



Chain-of-Thought

Reasoning

- ✓ *Step 1:* Use the formula for the area of a triangle, which is (1/2 * base * height).
- ✓ *Step 2:* *The* perpendicular from the top vertex splits the base into two segments of length 5 each.
- ✓ *Step 3:* Use the Pythagorean theorem for one of these right-angled triangles: h^2 + 5^2 = 13^2, h^2 = 13^2 - 5^2, h^2 = 169 - 25, h^2 = 144. So, h = √144 = 12.
- ✓ *Step 4:* Find the isosceles triangle's area: Area = (1/2 * base * height) = (1/2 * 10 * 12) = 60.
- ✓ *Step 5:* The area of the isosceles triangle is 60.

Ziyu G. et al. "Can We Generate Images with CoT? Let's Verify and Reinforce Image Generation Step by Step." CVPR 2025.

背景介绍

图像在推理前被转化为静态、固定的 Token 序列，无法持续进行交互



How many Calories does this drink contain?



Visual Tokens

Textual Tokens

背景介绍



How many calories does this drink contain?

[v]

One-time Encoding

Lossy

Image Caption (with error)

A Coca-Cola ad with the slogan "It's the real thing" appears on the left. On the right, **a nutrition label states 60 calories** per 12 oz serving.

图像向视觉 Token 的离散化转化是有损的，

细粒度信息的流失使模型在处理复杂任务时，

缺乏足够的视觉证据支撑

<thought> An advertisement for Coca-Cola is shown, including a bottle and a nutrition label. The label appears to show that the drink contains **60 calories** <thought>

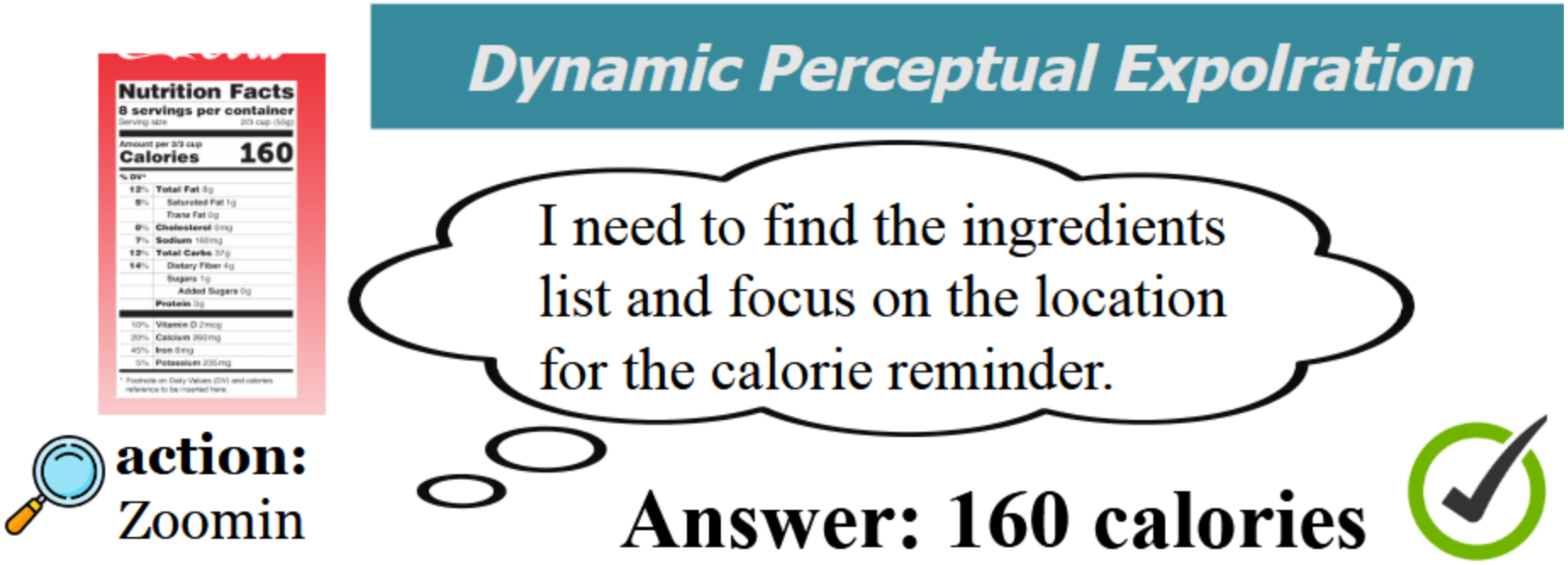
<answer> **It contains 60 calories** <answer>

Incorrect

背景介绍

一种解决方法：模型主动放大图像获取局部细节证据，

修正对初始图像的模糊感知

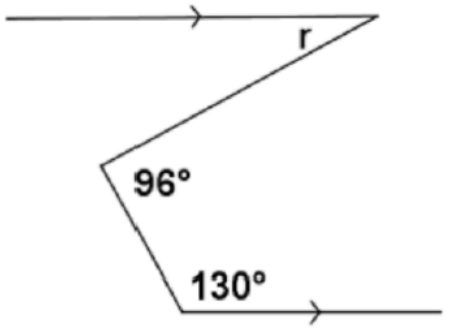


背景介绍

除了局部放大，还可以通过绘制辅助线等手段改写图像

将单纯的图像输入变成可以修改的草稿

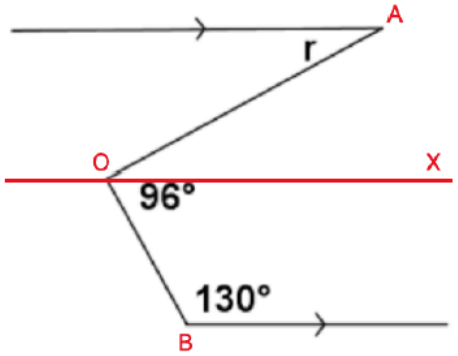
Structured Visual Reasoning



Draw **parallel auxiliary line OX through O**.
Use line properties to find angle r.

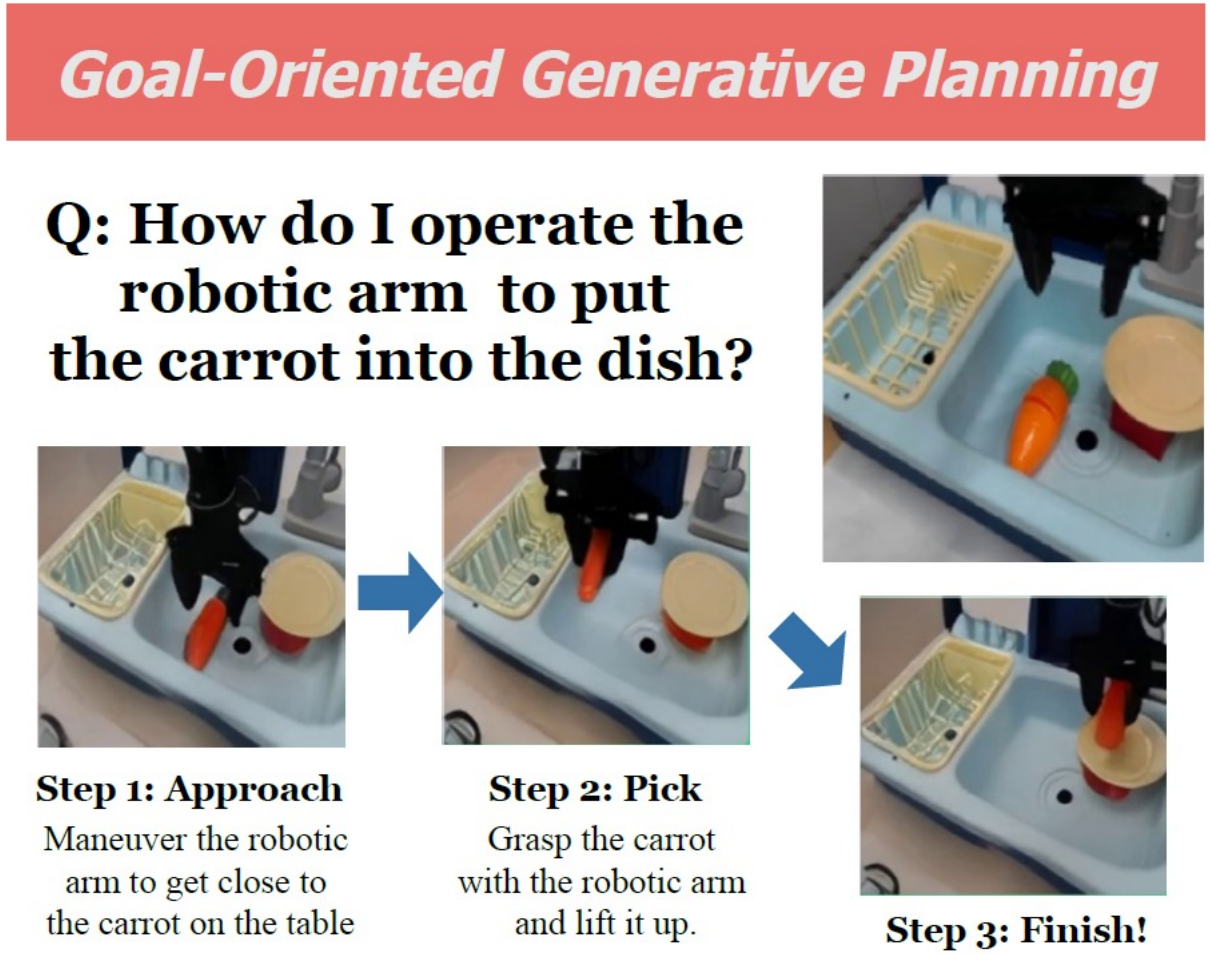
Answer: r = 46°

Q: Calculate the value of angle r



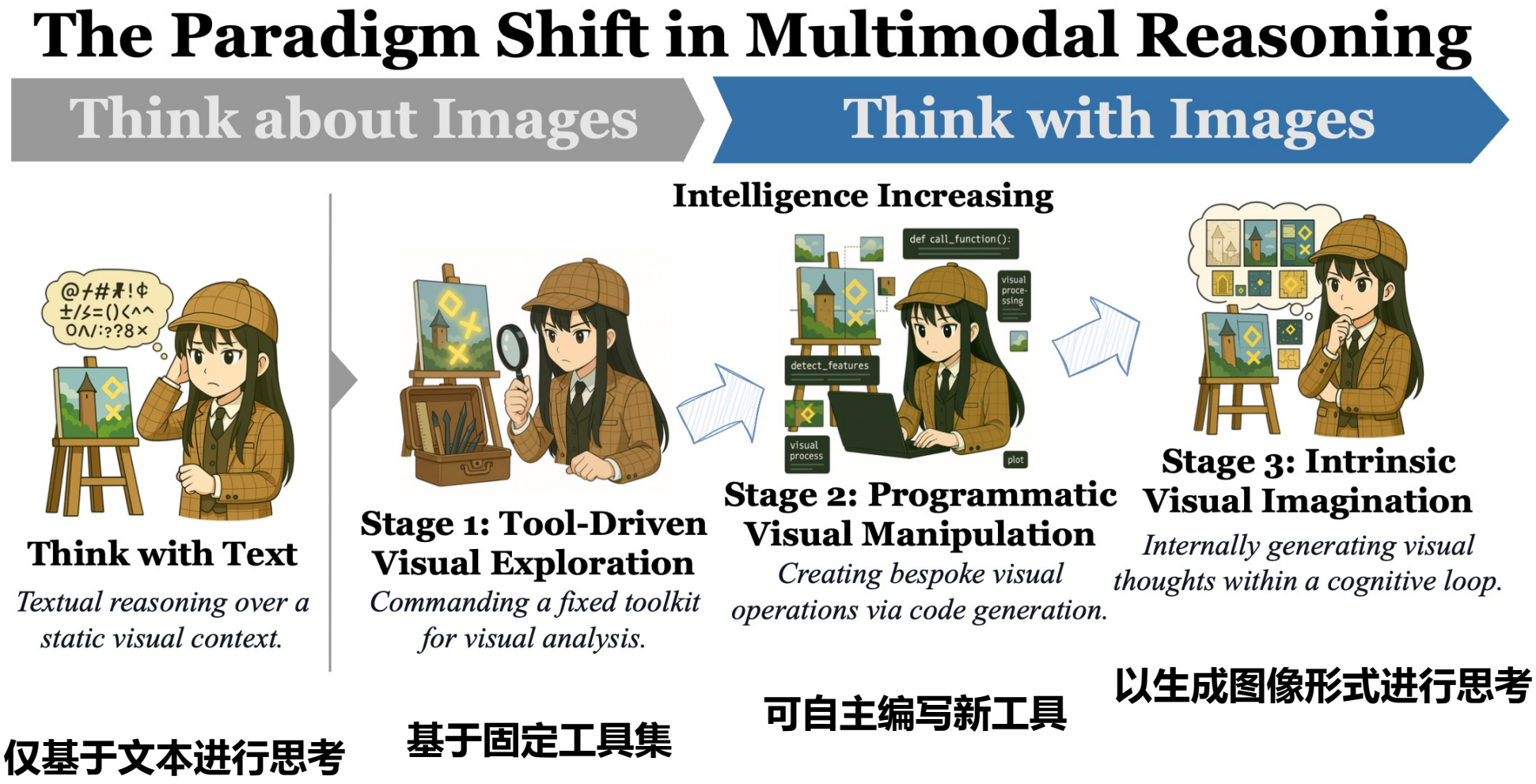
action:
draw auxiliary line

更高级的形式：生成“未来画面”模拟未来实现规划



背景介绍

随着推理范式的演进，模型对图像的操作能力更强，智能水平更高



仅基于文本进行思考

基于固定工具集

可自主编写新工具

以生成图像形式进行思考

课程提纲

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一、课题背景介绍

二、工具驱动的视觉探索

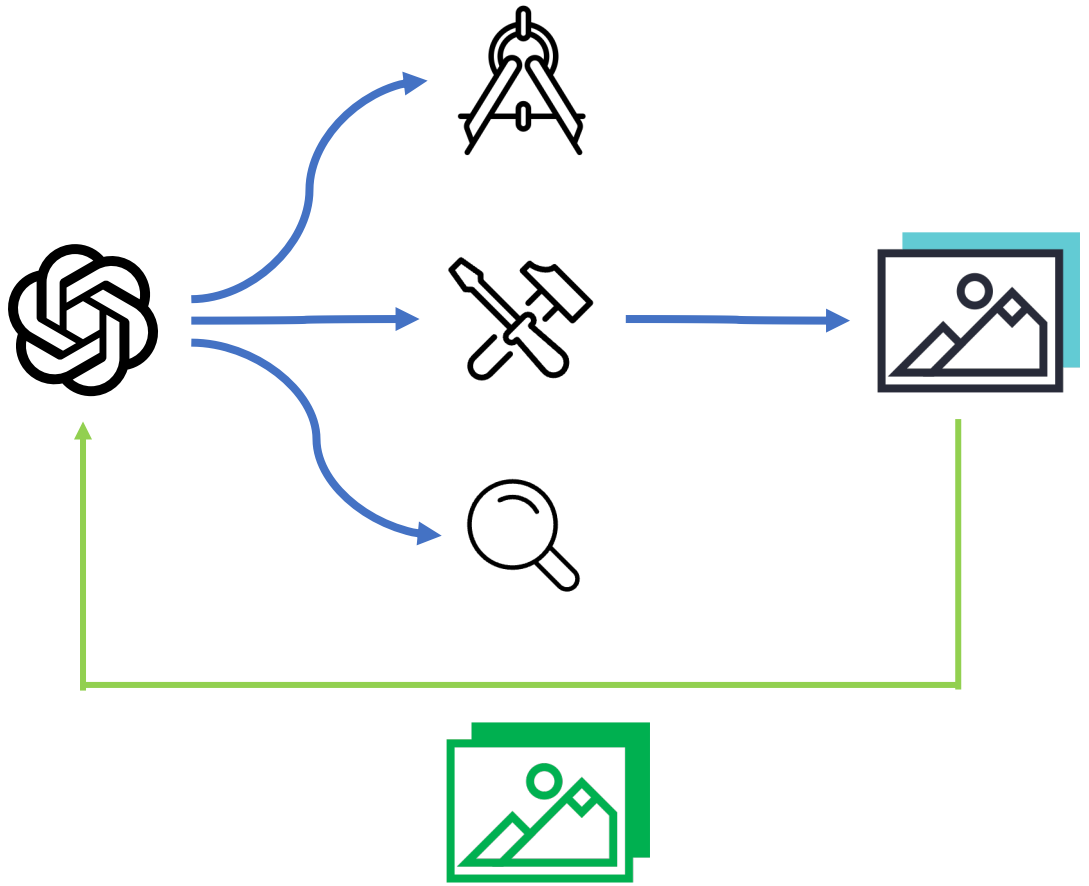
三、程序化视觉操作

四、原生视觉想象

五、总结与展望

工具驱动的视觉探索

阶段一：模型使用预先定义的固定工具集与图像进行交互



工具驱动的视觉探索

Initial Visual Tokens ... Initial Textual Tokens

工具调用前的上下文



Tool Call



将工具调用结果添加到上下文

Initial Visual Tokens ... Initial Textual Tokens ... New Visual Tokens

Answer:
160 Calories

工具驱动的视觉探索

模型可以调用的视觉工具种类是多样的



Stage (i): Tool-Driven Visual Exploration

① Semantic Grounding Tools



OCR

"Einstein Kit Best for Learning: Teaches over 100+ scientific concepts Age Appropriate: Just right for 6-12 year old children..."

Q: Is Einstein kit suitable for my 11 year old daughter?

A: Yes. Kit is very suitable for 11 year old children

② Perception Exploration Tools



Q: What type of tree is the highest in the picture?

Zoom in

Thoughts:
I need to see the details to identify it. Zooming in.



A: Coconut tree.

③ Visual Reasoning-Aid Tools



Q: How many pears are in the plate?

Point

Thoughts:
I'll mark the pears to keep track while counting.

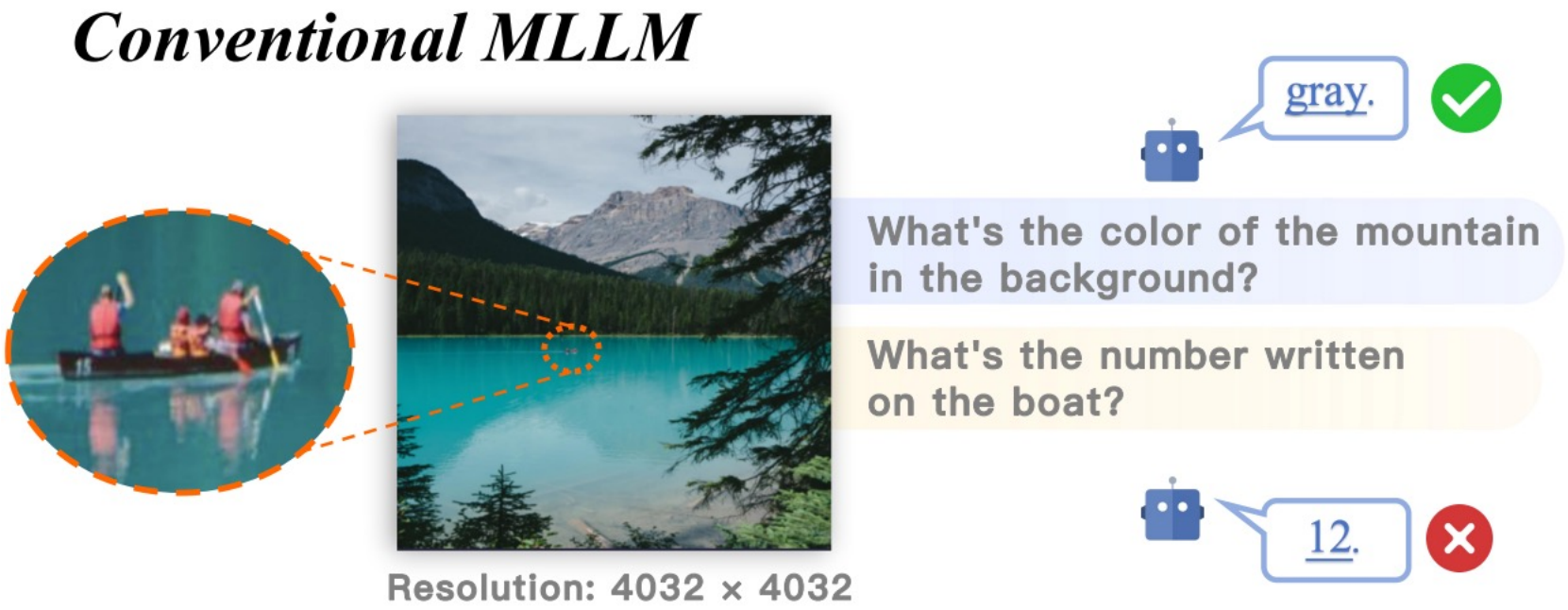


A: Three.

工具驱动的视觉探索 – ZoomEye

处理高分辨率图像时，多模态大模型可以有效识别主要物体，

却难以感知占据画面比例较小的细节信息



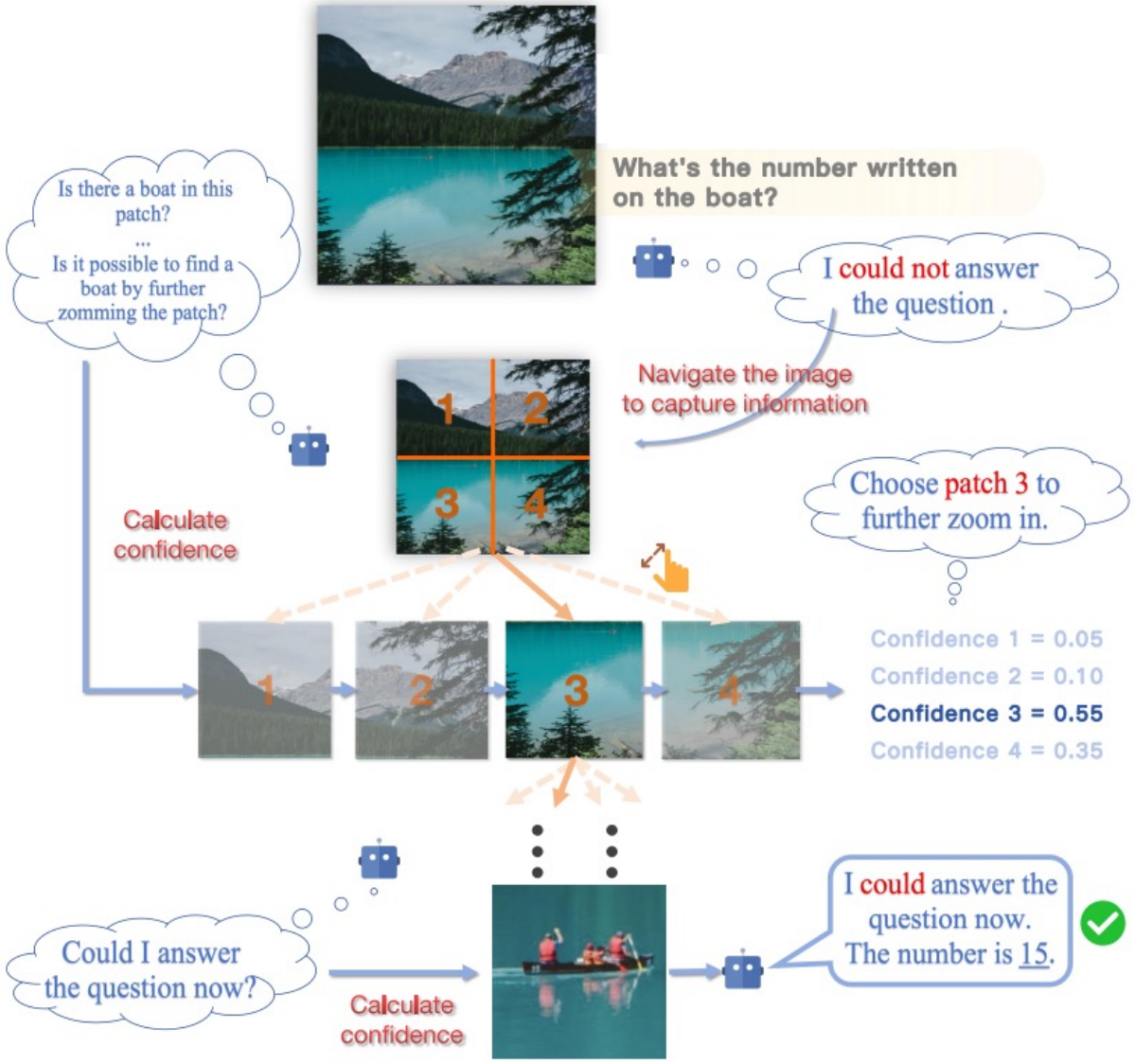
Haozhan S. et al. “ZoomEye: Enhancing Multimodal LLMs with Human-Like Zooming Capabilities through Tree-Based Image Exploration.” EMNLP 2025.

工具驱动的视觉探索 – ZoomEye

论文方法——树搜索：

当模型认为无法回答问题时，
将图像分为四个区域，
计算模型对每个区域的置信度，
选择置信度最大的区域放大，
重复直至模型足够自信回答问题

若没有找到期望信息，则回到上一级



Haozhan S. et al. "ZoomEye: Enhancing Multimodal LLMs with Human-Like Zooming Capabilities through Tree-Based Image Exploration." EMNLP 2025.

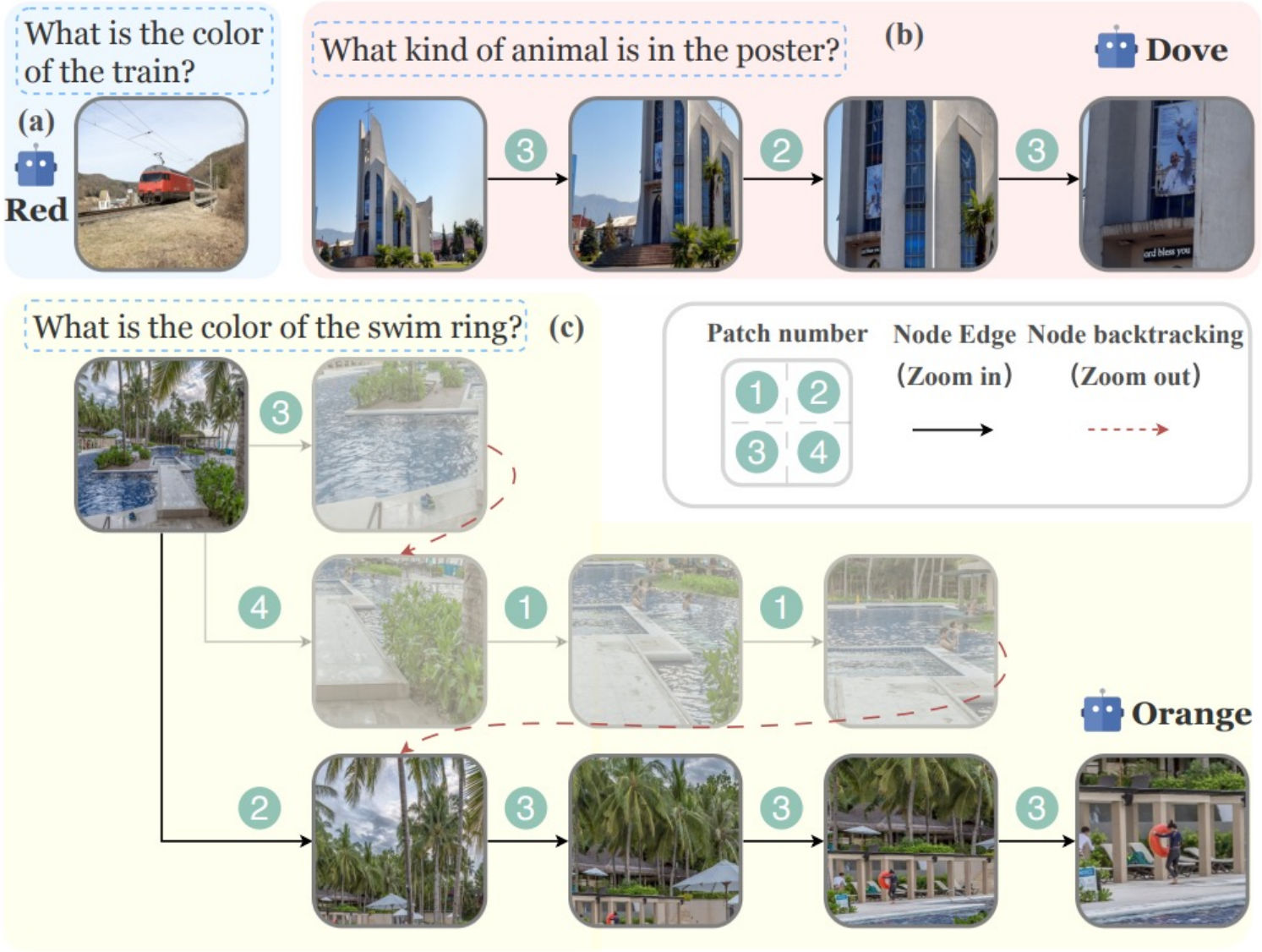
工具驱动的视觉探索 – ZoomEye

模型相当于具备两种“工具”：

<u>放大</u> 和 <u>缩小</u>

模型可以：

- **直接回答问题**
- **逐级放大获取细节信息**
- **缩小回到上一级，转而搜索其他路径**



Haozhan S. et al. “ZoomEye: Enhancing Multimodal LLMs with Human-Like Zooming Capabilities through Tree-Based Image Exploration.” EMNLP 2025.

工具驱动的视觉探索 – ZoomEye

ZoomEye 为 prompt-based 的 training-free 框架，

由 prompt 和代码引导模型的行为

```
prompts = {
    "global":{
        "pre_information": "<image>\n",
        "latent_prompt": "According to your common sense knowledge and the content of image, \
                          is it possible to find a {} by further zooming in the image? \
                          Answer Yes or No and tell the reason.",
        "existence_prompt": "Is there a {} in the image? Answer Yes or No.",
        "answering_prompt": "Question: {}\nCould you answer the question based on the the available visual information?\
                             Answer Yes or No.",
    },
    "zoom":{
        "pre_information": f"<image>\nThis is the main image, and the section enclosed by the red rectangle is the focus region.\
                            \n<image>\nThis is the zoomed-in view of the focus region.\n",
        "latent_prompt": "According to your common sense knowledge and the content of the zoomed-in view, along with its location in the image, \
                          is it possible to find a {} by further zooming in the current view? Answer Yes or No and tell the reason.",
        "existence_prompt": "Is there a {} in the zoomed-in view? Answer Yes or No.",
        "answering_prompt": "Question: {}\nCould you answer the question based on the the available visual information? Answer Yes or No.",
    },
}
```

Haozhan S. et al. “ZoomEye: Enhancing Multimodal LLMs with Human-Like Zooming Capabilities through Tree-Based Image Exploration.” EMNLP 2025.

工具驱动的视觉探索 – ZoomEye

置信度的计算方法：

- **存在置信度 c_e：询问 Is there a {} in the image? 模型回答 "Yes" 和 "No" 的概率比**
- **潜在置信度 c_l：询问 Is it possible to find a {} by further zooming? 模型回答 "Yes" 和 "No" 的概率比**
- **图像区域的优先级为两者的加权和：$(1-\alpha)c_e+\alpha c_l$，权重 α 由深度决定。**

function LOGITS RATIO(n_t, **x**)
$z_1 \leftarrow \Phi_\theta(y = \texttt{Yes} \mid \mathcal{V}(n_t),\ \mathbf{x})$
$z_2 \leftarrow \Phi_\theta(y = \texttt{No} \mid \mathcal{V}(n_t),\ \mathbf{x})$
$z \leftarrow (softmax(z_1, z_2)[0] - 0.5) \times 2)$
return z ▷ $z \in (-1, 1)$

function GET PRIORITY(n_t)
if n_t.priority is None **then**
$c_e \leftarrow$ LOGITS RATIO($n_t, \mathrm{p}_e(o)$)
$c_l \leftarrow$ LOGITS RATIO($n_t, \mathrm{p}_l(o)$)
$\alpha \leftarrow \mathcal{W}(n_t.\texttt{depth})$ ▷ weighted factor
$n_t.\texttt{priority} \leftarrow \alpha \cdot c_l + (1-\alpha) \cdot c_e$
return n_t.priority

- **回答置信度 c_a：询问 Could you answer the question based on the available visual information? 模型回答 "Yes" 和 "No" 的概率比**
  - **当 c_a 大于阈值时，停止搜索并得出答案。**

function $\mathcal{S}(n_t)$ ▷ Stopping Criterion
$c_a \leftarrow$ LOGITS RATIO($n_t, \mathrm{p}_a(q_s)$)
return $c_a \geq \tau$

Haozhan S. et al. "ZoomEye: Enhancing Multimodal LLMs with Human-Like Zooming Capabilities through Tree-Based Image Exploration." EMNLP 2025.

工具驱动的视觉探索 – ZoomEye

开源模型使用 ZoomEye 算法进行推理后，

在多个高分辨率图像问答评测基准上取得了较大的提升

Model	V^* Bench			HR-Bench 4K			HR-Bench 8K		
	Attr	Spatial	Overall	FSP	FCP	Overall	FSP	FCP	Overall
Closed-source MLLMs									
QWen-VL-max (Bai et al., 2023)	-	-	-	65.0	52.0	58.50	54.0	51.0	52.50
GPT4o (Achiam et al., 2023)	-	-	66.0	70.0	48.0	59.0	62.0	49.0	55.5
Baseline and Local Input Zoom Eye									
LLaVA-v1.5-7B (Liu et al., 2024a)	43.47	56.57	48.68	38.5	33.75	36.13	33.0	31.25	32.13
LLaVA-v1.5-7B w/ Zoom Eye	83.45	82.89	83.25	67.75	38.75	53.25	65.50	36.0	50.75
Δ	+40.48	+26.32	+34.57	+29.25	+5.0	+17.12	+32.50	+4.75	+18.62
Baseline and Global+Local Input Zoom Eye									
Qwen2.5VL-3B (Bai et al., 2025)	80.87	71.05	76.96	82.75	49.0	65.88	80.5	45.25	62.88
Qwen2.5VL-3B w/ Zoom Eye	88.70	**89.47**	89.01	86.75	53.50	70.13	84.75	52.0	68.38
Δ	+7.83	+18.42	+12.05	+4.0	+4.50	+4.25	+4.25	+6.75	+5.50
LLaVA-ov-7B (Li et al., 2024)	75.65	75.0	75.39	72.0	54.0	63.0	67.25	52.25	59.75
LLaVA-ov-7B w/ Zoom Eye	**93.91**	85.53	**90.58**	84.25	55.0	69.63	88.5	50.0	69.25
Δ	+18.26	+10.53	+14.19	+12.25	+1.0	+6.63	+21.25	-2.25	+10.0
InternVL2.5-8B (Chen et al., 2024a)	67.83	71.05	69.11	75.75	56.25	66.0	61.5	53.25	57.38
InternVL2.5-8B w/ Zoom Eye	86.09	82.89	84.82	**88.75**	**61.50**	**75.13**	**89.75**	**57.5**	**73.63**
Δ	+18.26	+11.84	+15.71	+13.0	+5.25	+9.13	+28.25	+4.25	+16.25

Haozhan S. et al. “ZoomEye: Enhancing Multimodal LLMs with Human-Like Zooming Capabilities through Tree-Based Image Exploration.” EMNLP 2025.

工具驱动的视觉探索 – VLM-R³

依靠纯文本推理的多模态模型难以处理细粒度的视觉任务

- **令模型学会对图像进行裁剪放大**



What is the total volume of the measuring cup?

{"bbox_2d": [616, 212, 1021, 599]}



<answer>1000 ml</answer>

Chaoya J. et al. "VLM-R³: Region Recognition, Reasoning, and Refinement for Enhanced Multimodal Chain-of-Thought." NeurIPS 2025.

工具驱动的视觉探索 – VLM-R³

仅使用 裁剪/crop 工具聚焦图像的特定区域

```
You need to first think about the reasoning process in your mind, and then
provide the answer. When thinking, you should call the "crop" tool (format:
{"bbox_2d": [x1, y1, x2, y2]}) to focus on the key areas in the image. The
reasoning process and the answer are included in the <think> </think> and
<answer> </answer> tags respectively.
```

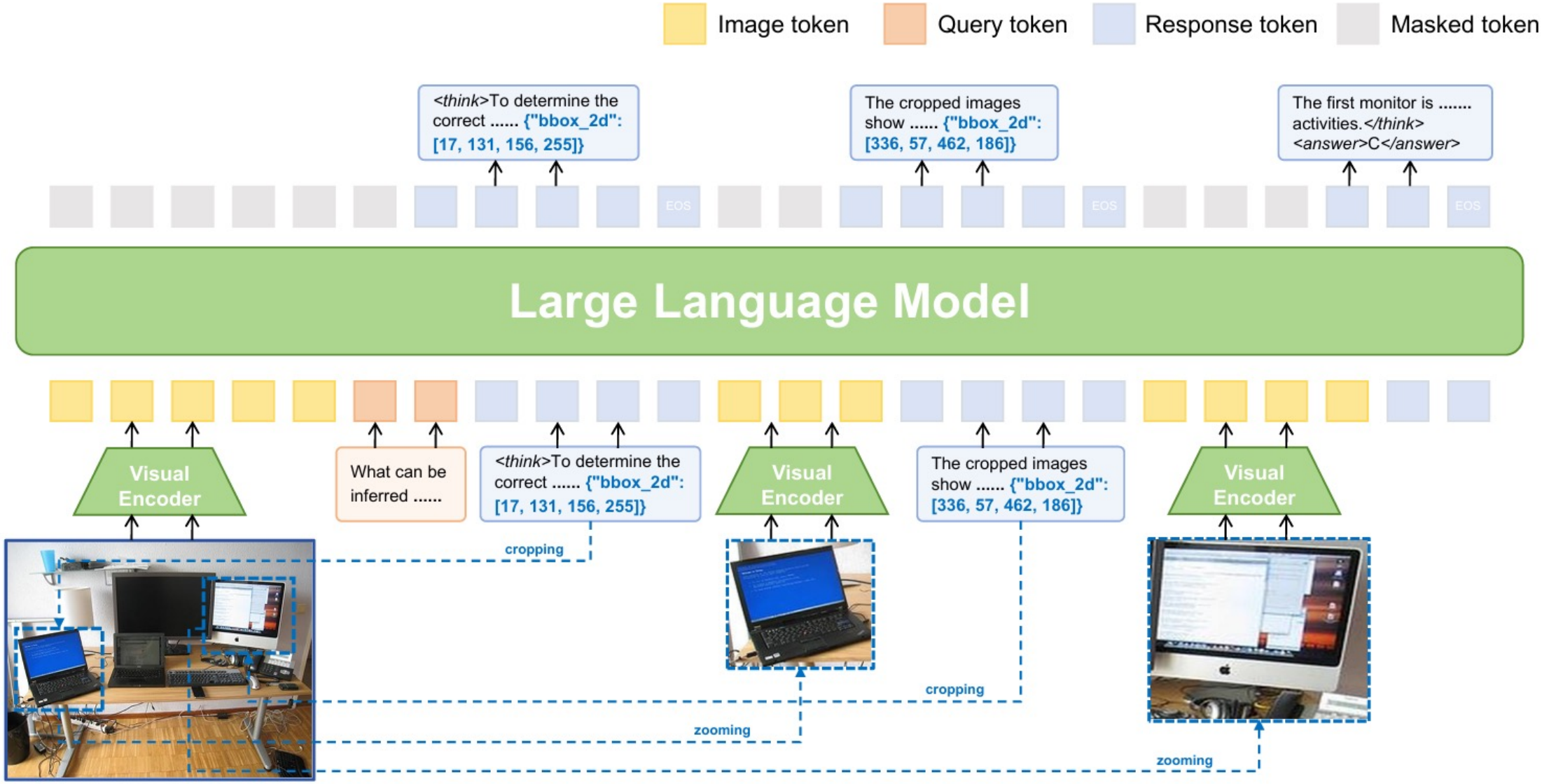
To determine the total volume of the measuring cup, I need to focus on the markings on the side of the cup. The markings indicate the volume in milliliters (ml) and grams (g). The maximum volume is clearly marked at the top of the cup.
Step 1: Use the Crop tool to zoom in on the markings on the measuring cup to ensure accuracy.
{"bbox_2d": [616, 212, 1021, 599]}



Step 2: Analyze the markings. The maximum volume is indicated by the highest number on the scale, which is 1000 ml.
</think>
<answer>1000 ml</answer>

Chaoya J. et al. "VLM-R³: Region Recognition, Reasoning, and Refinement for Enhanced Multimodal Chain-of-Thought." NeurIPS 2025.

工具驱动的视觉探索 – VLM-R³



Chaoya J. et al. "VLM-R³: Region Recognition, Reasoning, and Refinement for Enhanced Multimodal Chain-of-Thought." NeurIPS 2025.

工具驱动的视觉探索 – VLM-R³

两阶段训练：有监督微调 (SFT) + 强化学习 (RL)

利用大模型构建 SFT 阶段所需的数据集 VILR

TextVQA

DocVQA

GQA

从多个现有数据集
选取数据

利用 Qwen2.5-VL 72B,
GPT 等模型
生成图像-文本交错推理链

过滤低质量数据：

- 用 VLM 验证边界框是否包含完整有意义的视觉单元
- 用 DeepSeek-V3 验证文本推理步骤是否逻辑连贯

最终得到 12K 训练数据

Chaoya J. et al. “VLM-R³: Region Recognition, Reasoning, and Refinement for Enhanced Multimodal Chain-of-Thought.” NeurIPS 2025.

工具驱动的视觉探索 – VLM-R³

第二阶段 GRPO 强化学习奖励设计：

- **Accuracy Reward：** 答对问题给予奖励 1，否则奖励值为 0;
- **Format Adherence Reward：** 回答格式正确给予奖励 1;
- **Region Validity Reward：** 生成符合格式且不重复的 {"bbox_2d": [x1, y1, x2, y2]} 工具调用格式时，给予奖励 0.5;
- **Reasoning Length Reward：**
  <think></think> 中的推理文本，每一个字符给予 0.001 奖励，最大奖励限制在 0.25

Chaoya J. et al. "VLM-R³: Region Recognition, Reasoning, and Refinement for Enhanced Multimodal Chain-of-Thought." NeurIPS 2025.

工具驱动的视觉探索 – VLM-R³

与基础模型 Qwen2.5-VL 7B 相比，

经过完整训练的模型在不同评测基准的分数更高

Model Variant	MathVista	MMMU	ScienceQA	DocVQA	Avg.
Base Model (Qwen2.5-VL)	68.2	58.6	73.6	95.7	74.0
w/o Interleaved Chain-of-Thought	67.1 (↓3.3)	59.4 (↓2.8)	75.4 (↓12.5)	95.9 (↓0.9)	74.4 (↓4.9)
w/o VLIR Fine-tuning	65.8 (↓4.6)	57.0 (↓5.2)	72.2 (↓15.7)	93.3 (↓3.5)	72.1 (↓7.2)
w/o R-GRPO	69.7 (↓0.7)	60.8 (↓1.4)	84.6 (↓3.3)	96.1 (↓0.7)	77.8 (↓1.5)
Full VLM-R^3 (Ours)	**70.4**	**62.2**	**87.9**	**96.8**	**79.3**

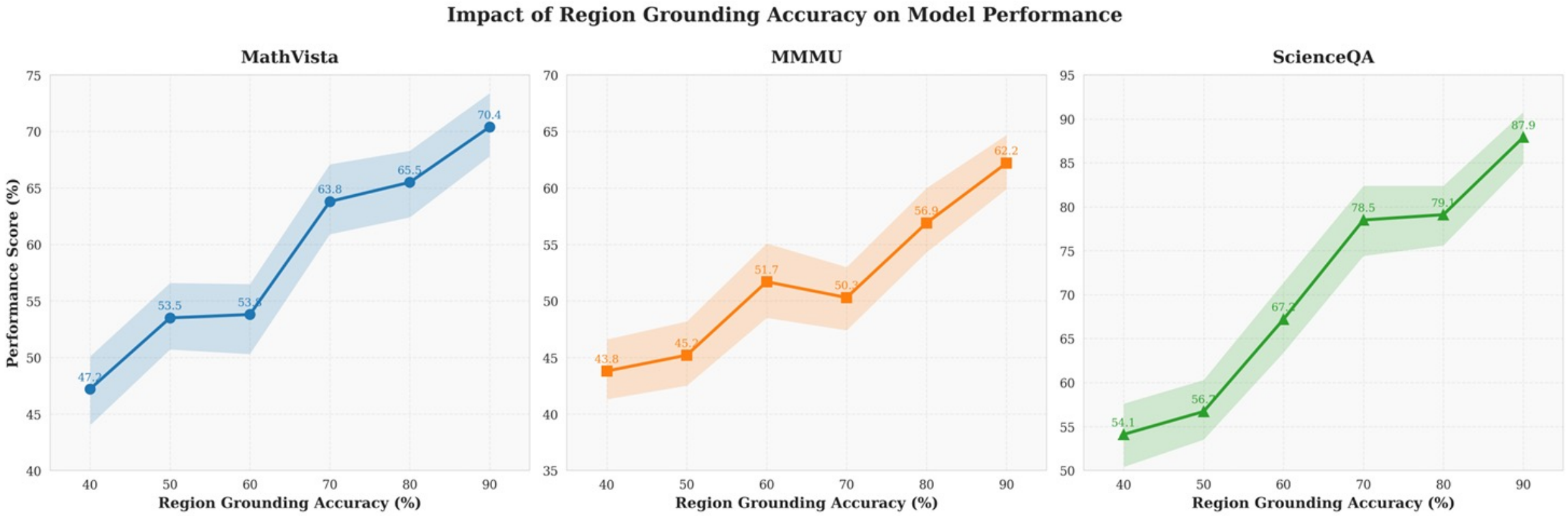
去掉 图文交错思维链/SFT/RL 其中一个组件都会导致性能下降

Chaoya J. et al. “VLM-R³: Region Recognition, Reasoning, and Refinement for Enhanced Multimodal Chain-of-Thought.” NeurIPS 2025.

工具驱动的视觉探索 – VLM-R³

剪裁区域的准确率越高，回答问题的准确率也越高

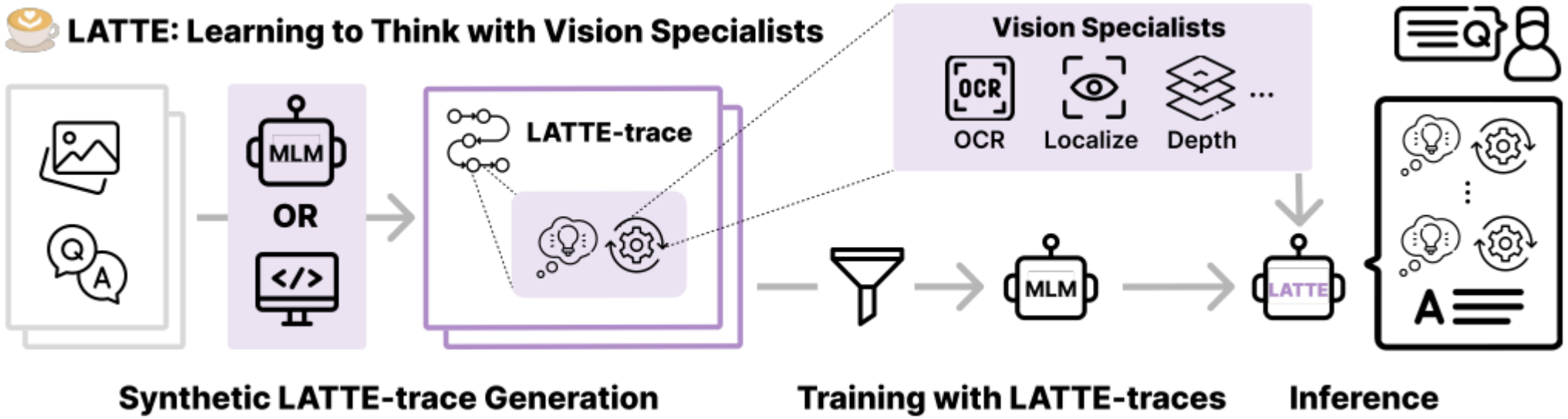
说明了选择正确视觉证据的重要性



Chaoya J. et al. “VLM-R³: Region Recognition, Reasoning, and Refinement for Enhanced Multimodal Chain-of-Thought.” NeurIPS 2025.

工具驱动的视觉探索 – LATTE

更多工具：OCR, 物体检测，深度估计……



Zixian M. et al. “LATTE: Learning to Reason with Vision Specialists.” CVPR 2025 Workshop SynData4CV.

工具驱动的视觉探索 – LATTE



Q: How many gallons of supreme gasoline can I get with $50?
A: 13.7

Mantis-LLaVA-7B: 3.6
LLaVA-OV: 5.2

LATTE

t_0: I need to extract the price of supreme gasoline from the image.
a_0: **OCR**(image-0)
o_0: "Gasoline - Self Serve - Regular - 3.49 9/10 - Plus - 3.55 9/10 - Supreme - 3.65 9/10 - Diesel No. 2 - 3.60 9/10"

t_1: The price of supreme gasoline is $3.65 per gallon...I need to divide $50 by $3.65.
a_1: **Calculate**("50 / 3.65")
o_1: 13.6986

t_2: The calculation shows that you can get approximately 13.7 gallons of supreme gasoline with $50.
a_2: **Terminate**("13.7")



Q: Can the boy reach the highest book?
(A) Yes, (B) No
A: (B)

Mantis-LLaVA: (A)
LLaVA-OV: A. The boy can reach the book.

LATTE

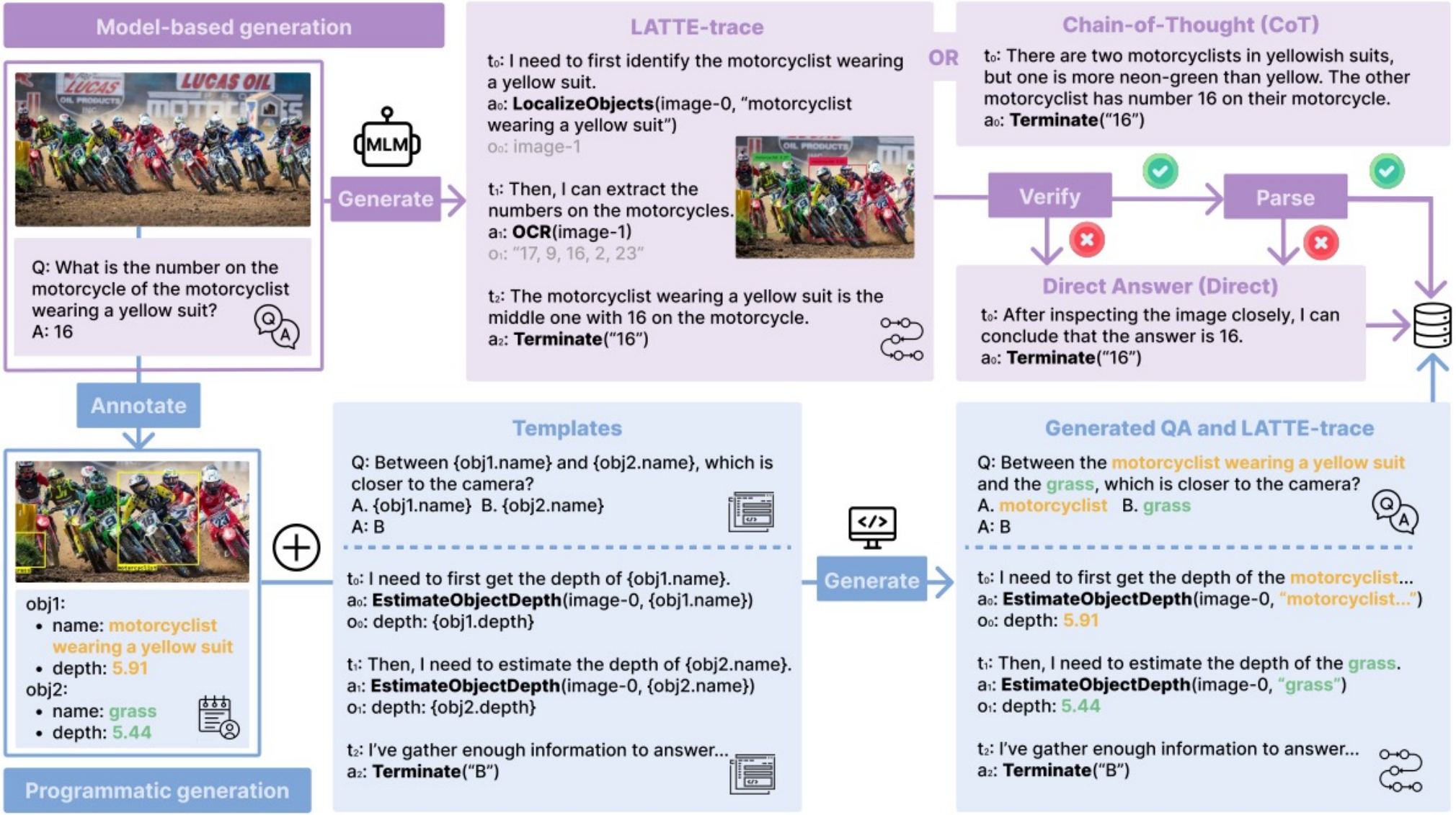
t_0: To determine if the boy can reach the highest book, I need to analyze the image for the boy's height and the height of the bookshelf.
a_0: **LocalizeObjects**(image-0, ["boy", "bookshelf"])
o_0: [{'label': 'boy', 'bbox': [0.07, 0.48, 0.18, 0.95], 'score': 0.89},
{'label': 'bookshelf', 'bbox': [0.13, 0.01, 0.32, 0.72], 'score': 0.72}]



t_1: The boy's height is below the bookshelf, indicating he cannot reach the highest book.
a_1: **Terminate**("B")

Zixian M. et al. "LATTE: Learning to Reason with Vision Specialists." CVPR 2025 Workshop SynData4CV.

工具驱动的视觉探索 – LATTE



Zixian M. et al. "LATTE: Learning to Reason with Vision Specialists." CVPR 2025 Workshop SynData4CV.

工具驱动的视觉探索 – LATTE

和只用问题+答案进行训练相比，

使用调用工具推理轨迹数据训练能提高模型的表现

Language / Vision	Starting checkpoint	Method	Perception				Perception + Reasoning				Overall
			CV-Bench	BLINK	RealWorldQA	Avg	MathVista	MMStar	MMVet	Avg	Avg
LLaMA3-8B / CLIP	Mantis Pretrained	Vanilla IT	52.6	45.8	52.3	50.2	33.1	36.7	28.9	32.9	41.6
		LATTE	56.9	49.6	51.1	52.6	36.6	40.8	45.2	40.8	46.7 (+5.1)
LLaMA3-8B / SigLIP		Vanilla IT	52.3	43.7	51.8	49.3	31.1	40.5	33.0	34.9	42.1
		LATTE	57.2	47.8	53.7	52.9	34.9	44.6	45.2	41.6	47.2 (+5.1)
	Mantis Instruct-tuned	Vanilla IT	50.6	46.7	54.8	50.7	36.2	40.7	29.7	35.5	43.1
		LATTE	51.7	47.3	56.1	51.7	38.9	45.1	50.0	44.7	48.2 (+5.1)
Qwen2-7B / SigLIP	LLaVa-OV Stage 1.5	Vanilla IT	56.8	50.3	57.8	55.0	42.4	50.1	39.3	43.9	49.5
		LATTE	**60.2**	**52.6**	**61.1**	**58.0**	**46.9**	**50.8**	**50.9**	**51.2**	**53.8** (+4.3)

Zixian M. et al. “LATTE: Learning to Reason with Vision Specialists.” CVPR 2025 Workshop SynData4CV.

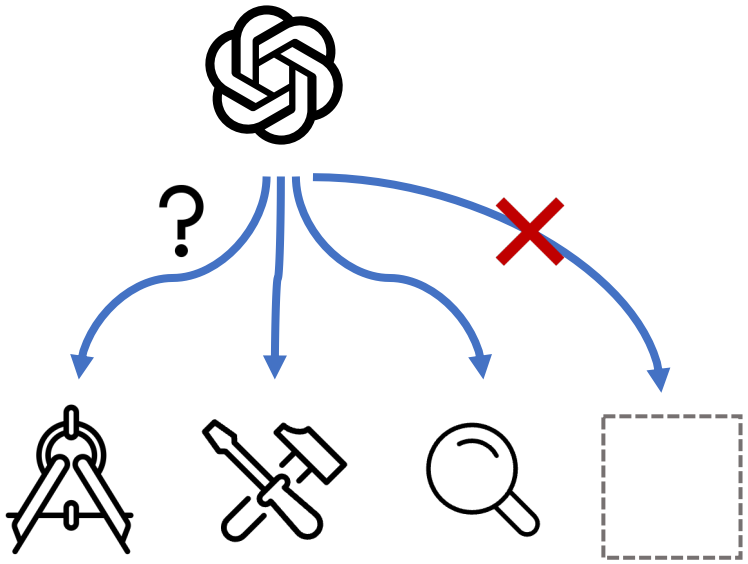
课程提纲

CONTENTS

一、课题背景介绍
二、工具驱动的视觉探索
三、程序化视觉操作
四、原生视觉想象
五、总结与展望

程序化视觉操作

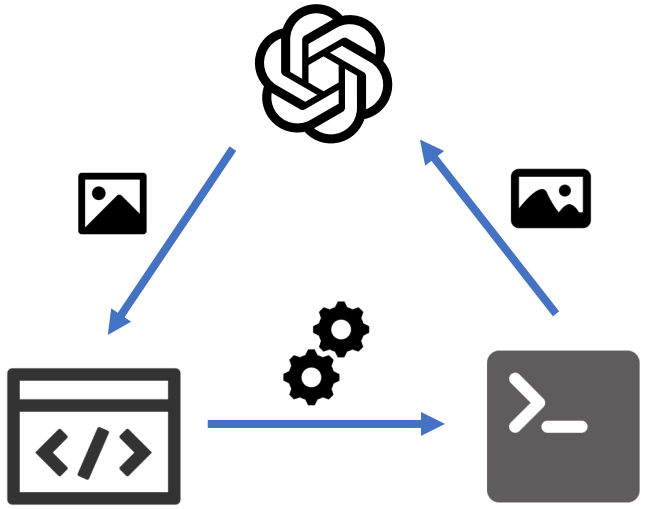
Stage (i): Tool-Driven Visual Exploration



固定、有限的工具集

模型只能从中选择

Stage (ii): Programmatic Visual Manipulation



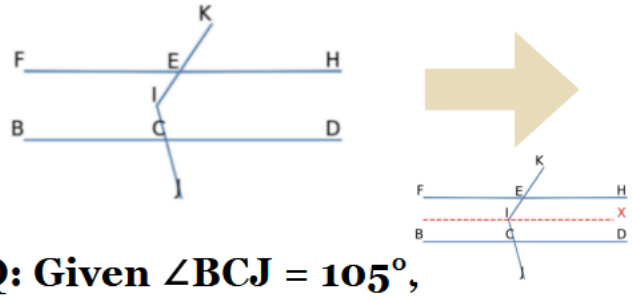
基于代码操作、生成工具

实现更灵活、复杂的视觉操作



Stage (ii): Programmatic Visual Manipulation

① Visual Operations



Q: Given ∠BCJ = 105°, ∠KEF = 105°, find ∠EIC.

Thoughts:

Draw an auxiliary line through point will help solve it

```
# Define auxiliary point X for line IX
x_I = points[“I”][0]
y_I = points["I"][1]
x_X = x_I + 100
y_X = y_I + slope_bd * (x_X - x_I)
ax.plot([x_I, x_X], [y_I, y_X], ‘red')
```

A: 150°

② Logical Operations



Q: Where is the bookshelf relative to the chair to the left of the vase?

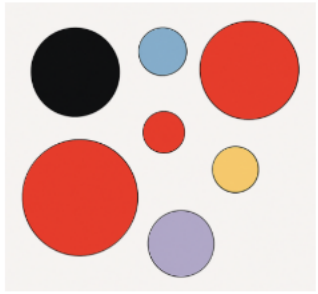
Thoughts:

Building the spatial logic for 'chair left of vase'

```
for v_patch in vases:
    chairs_on_left = [
        c_patch for c_patch in chairs
        if c_patch.horizontal.center <
v_patch.horizontal.center
    ]
```

A: It is to the left.

③ Composite Operations



Q: Find red circles larger than the average.



Thoughts:

I will Calculate average area, then filter the answer.

```
circles = find_all(image, "red & circle")
avg_area = average([c.area for c in circles])
large_red_circles = filter(circles,
    lambda c: c.area > avg_area)
mark_objects(image, large_red_circles)
```

A: Two circles.

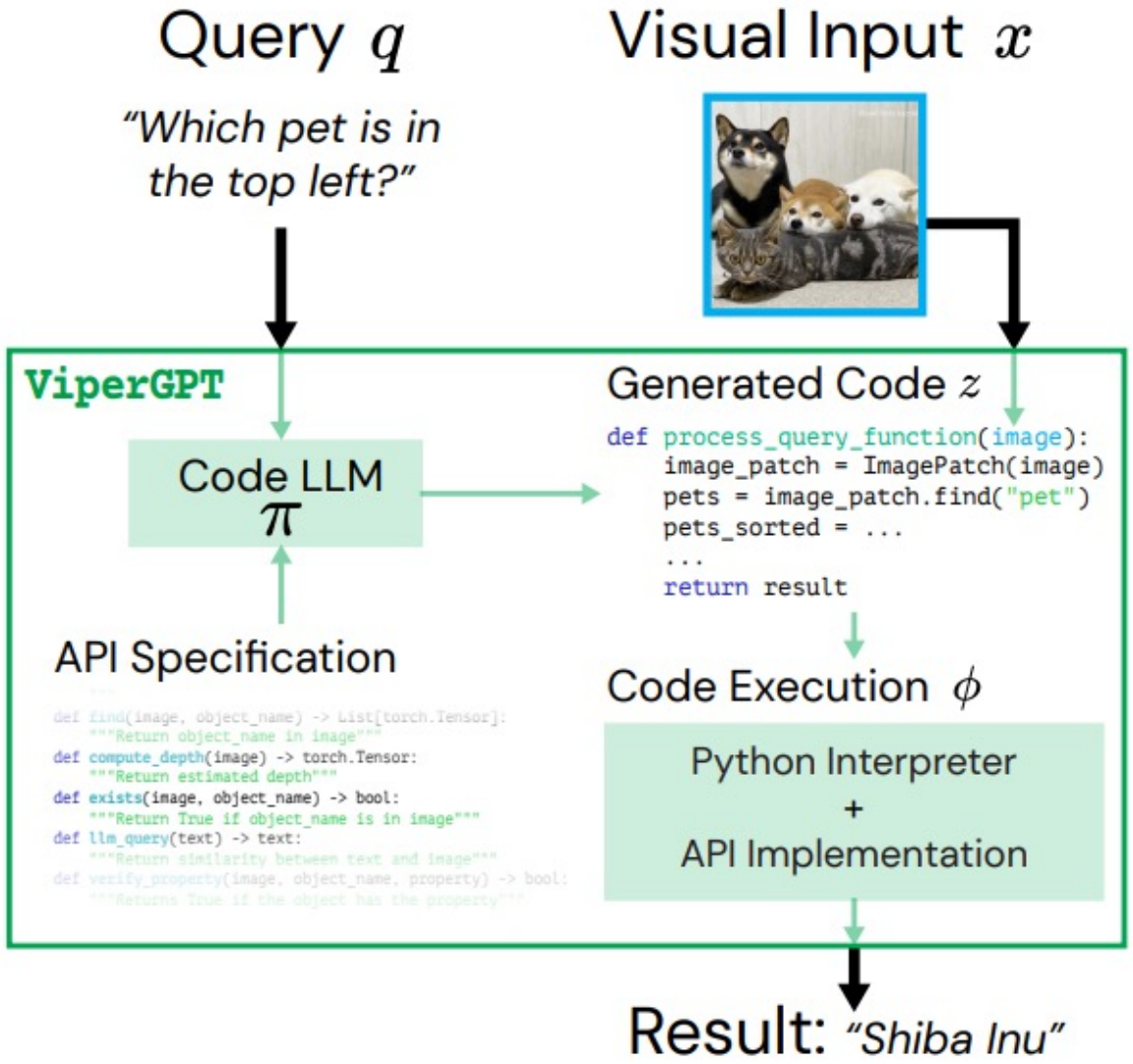
程序化视觉操作 – ViperGPT

核心观点：

视觉问答任务需要视觉处理和推理两种能力

——将视觉处理交给各类专用视觉模型，

LLM 负责推理，生成代码编排模型，总结答案



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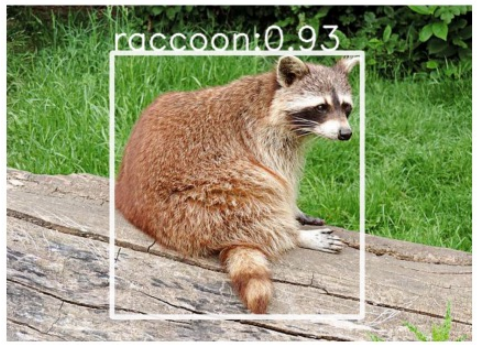
程序化视觉操作 – ViperGPT

Codex (code-davinci-002)

生成代码

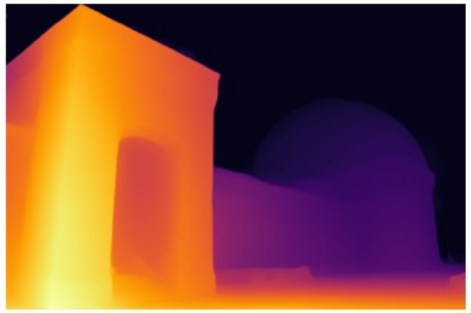
通过 API **间接**调用各类模型

API



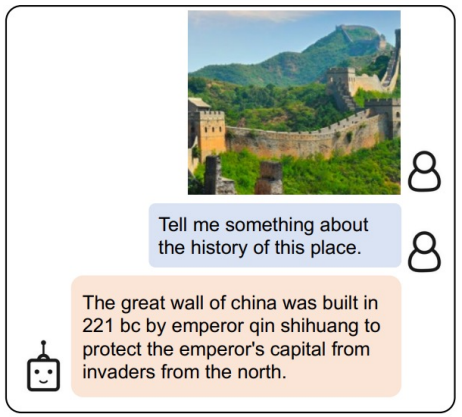
GLIP

目标检测



MiDaS

深度估计



BLIP-2

简单视觉问答



X-VLM

图文相似度

GPT-3

(text-davinci-003)

外部知识

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程序化视觉操作 – ViperGPT

将各类模型封装为各类 API，LLM 面向 API 编写代码

```
class ImagePatch:
    def __init__(self, image, left: int=None, lower: int=None, right: int=None, upper: int=None):

    def find(self, object_name: str) → List[ImagePatch]:  # GLIP 目标检测

    def exists(self, object_name: str) → bool:

    def verify_property(self, object_name: str, property: str) → bool:

    def best_text_match(self, option_list: List[str]) → str:  # X-VLM 图文相似度匹配

    def simple_query(self, question: str = None) → str:  # BLIP-2 视觉问答

    def compute_depth(self):  # MiDaS 深度估计

    def crop(self, left: int, lower: int, right: int, upper: int) → ImagePatch:

    def overlaps_with(self, left, lower, right, upper):

def best_image_match(list_patches: List[ImagePatch], content: List[str], return_index=False) → Union[ImagePatch, int]:

def distance(patch_a: ImagePatch, patch_b: ImagePatch) → float:

def bool_to_yesno(bool_answer: bool) → str:

def llm_query(question: str) → str:  # 调用 GPT-3 基于外部知识回答问题
```

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程序化视觉操作 – ViperGPT

Query: Drink with zero alcohol

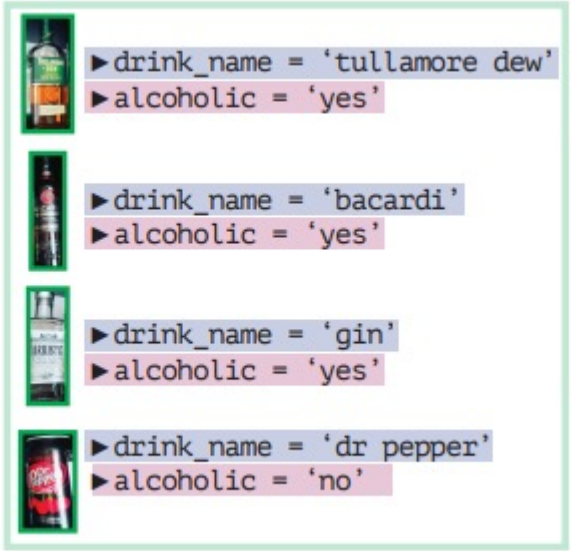


Generated Code

```
def execute_command(image):
    image_patch = ImagePatch(image)
    drink_patches = image_patch.find("drink")
    for drink_patch in drink_patches:
        drink_name = drink_patch.simple_query("What is this?")
        alcoholic = llm_query(f"Does the {drink_name} have alcohol?")
        if alcoholic == "no":
            return drink_patch
    return None
```

Execution





Result:



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程序化视觉操作 – ViperGPT

Query: pizza front

In:

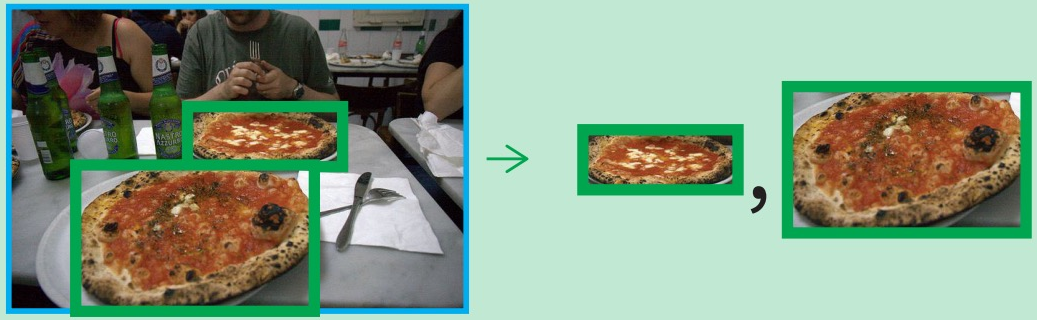


Generated Code

```
def execute_command(image):
    image_patch = ImagePatch(image)
    pizza_patches = image_patch.find("pizza")
    pizza_patches.sort(key=lambda pizza: pizza.compute_depth())
    patch_return = pizza_patches[0]
    return patch_return
```

Execution

```
pizza_patches = image_patch.find("pizza")
▶ pizza_patches= {List[ImagePatch]}
```



```
pizza.compute_depth()
```




```
pizza_patches.sort()
```



```
patch_return = pizza_patches[0]
return patch_return
```

Result:



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程序化视觉操作 – ViperGPT

Table 1. **RefCOCO Results**. We report accuracy on the REC task and testA split. ZS=zero shot, Sup.=supervised.

		IoU (%) ↑ RefCOCO	RefCOCO+
Sup.	MDETR [53]	90.4	85.5
	OFA [53]	94.0	91.7
ZS	OWL-ViT [38]	30.3	29.4
	GLIP [31]	55.0	52.2
	ReCLIP [49]	58.6	60.5
	ViperGPT (ours)	**72.0**	**67.0**

Table 2. **GQA Results**. We report accuracy on the test-dev set.

		Accuracy (%) ↑
Sup.	LGCN [20]	55.8
	LXMERT [51]	60.0
	NSM [24]	63.0
	CRF [39]	72.1
ZS	BLIP-2 [30]	44.7
	ViperGPT (ours)	**48.1**

在各个数据集上的表现
均优于直接使用专用视觉模型

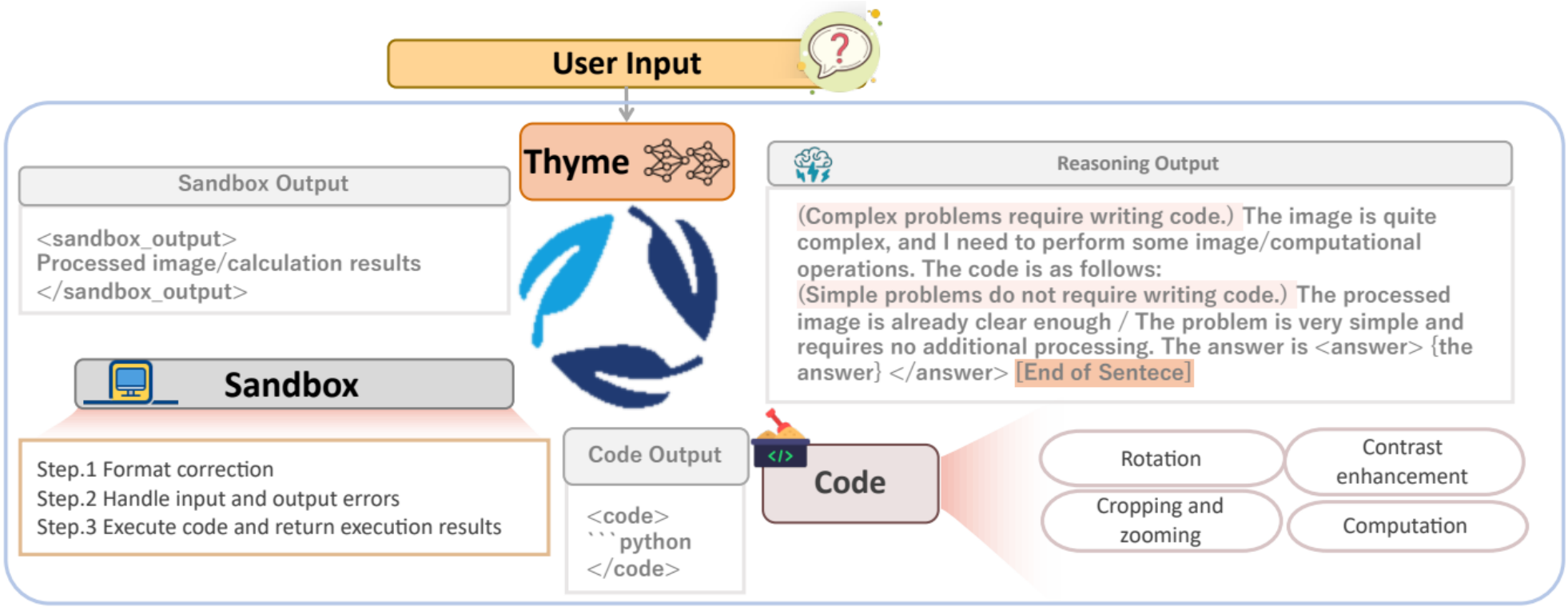
Table 3. **OK-VQA Results**.

		Accuracy (%) ↑
Sup.	TRiG [13]	50.5
	KAT [16]	54.4
	RA-VQA [32]	54.5
	REVIVE [33]	58.0
	PromptCap [21]	58.8
ZS	PNP-VQA [52]	35.9
	PICa [60]	43.3
	BLIP-2 [30]	45.9
	Flamingo [1]	50.6
	ViperGPT (ours)	**51.9**

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程序化视觉操作 – Thyme

- 基于 Qwen2.5-VL-7B 进行 SFT+RL 端到端训练
- 模型自主决定是否需要图像操作，简单问题可以直接回答
- 调用 OpenCV 库进行旋转、缩放等图像操作，不调用专用视觉模型

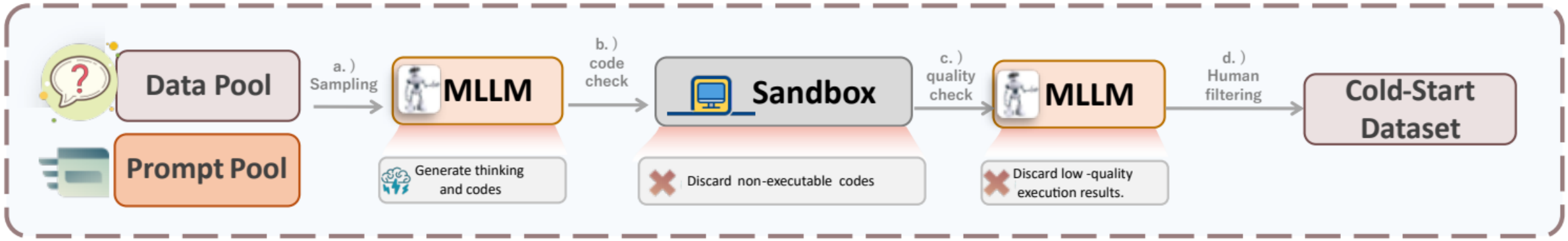


Yifan Z. et al. "Thyme: Think Beyond Images" ICLR 2026.

程序化视觉操作 – Thyme

SFT 数据构造流水线：

1. **从公开数据集中筛选出涵盖多种任务类型的 400 万个样本构成数据池；**
2. **Qwen2.5-VL-72B 基于数据池中的样本生成思考过程和执行代码；**
3. **在沙箱中运行生成的代码，丢弃包含不可执行代码的样本；**
4. **Qwen2.5-VL-72B 检查代码执行的结果是否符合预期，丢弃包含低质量执行结果的样本；**
5. **人工筛选，最后得到约 50 万个样本的 SFT 数据集。**

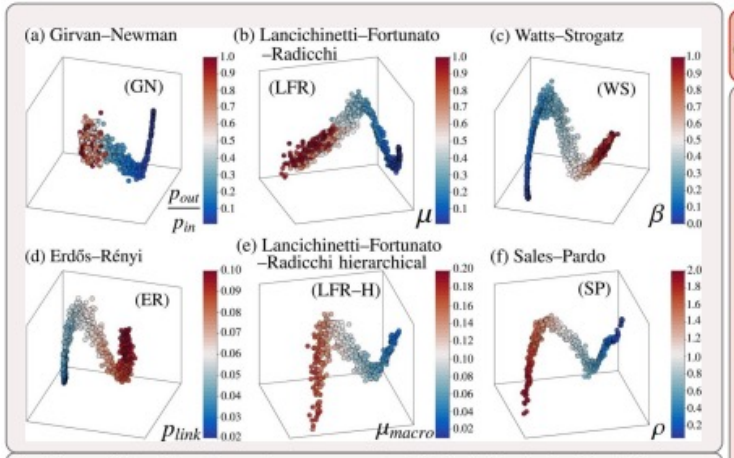


Yifan Z. et al. "Thyme: Think Beyond Images" ICLR 2026.

程序化视觉操作 – Thyme

SFT 数据示例，包含：

- **思考过程**
- **生成的代码**
- **代码的执行结果**
- **答案**



In figure (f), what does the parameter (\rho) likely indicate?
The choices are listed below:
A. The resilience of the network to attacks
B. The robustness of community detection algorithms
C. The assortativity coefficient of the network
D. The network's edge density

```python
import cv2

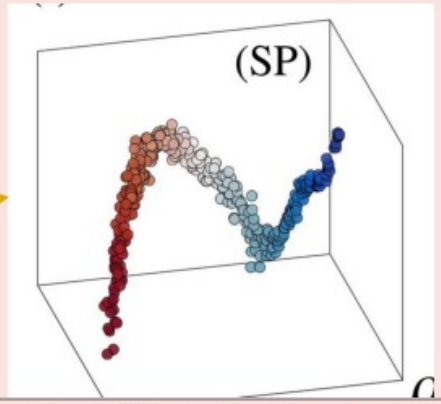
# Load the image
image_path = "/mnt/data/sci_chart.jpg"
image = cv2.imread(image_path)
# Define the coordinates for cropping figure (f)
x1, y1, x2, y2 = 1360, 700, 1900, 1200
# Crop the image
cropped_image = image[y1:y2, x1:x2]
# Apply zoom (resizing)
zoom_factor = 2
resized_image = cv2.resize(
    cropped_image, None, fx=zoom_factor,
    fy=zoom_factor, interpolation=cv2.INTER_LINEAR
)
...
# Print the path of the saved cropped image
print(processed_path)
```

Thinking

To address the question regarding the parameter (\rho) in figure (f), we need to examine the specific aspects of that figure closely. Cropping and zooming in on figure (f) will facilitate a better understanding of the parameter's significance within the network model. This task involves interpreting the role of the parameter (\rho) in the context of the Sales–Pardo model, and having a clearer view of the figure will aid significantly in this analysis.

Figure (f) is located in the bottom-right section of the image, and cropping this area will highlight the pertinent details, allowing for a more in-depth examination of the parameter (\rho).

Tool-call with code



<sandbox_output> **Returned Sub-image** <sandbox_output>

In figure (f), the parameter (\rho) is located on the axis where there is a noticeable shift in the density of the points. This suggests that (\rho) may be indicative of the edge density or the probability of edge creation within the Sales–Pardo model, significantly impacting the overall connectivity of the network.

<answer>D </answer>

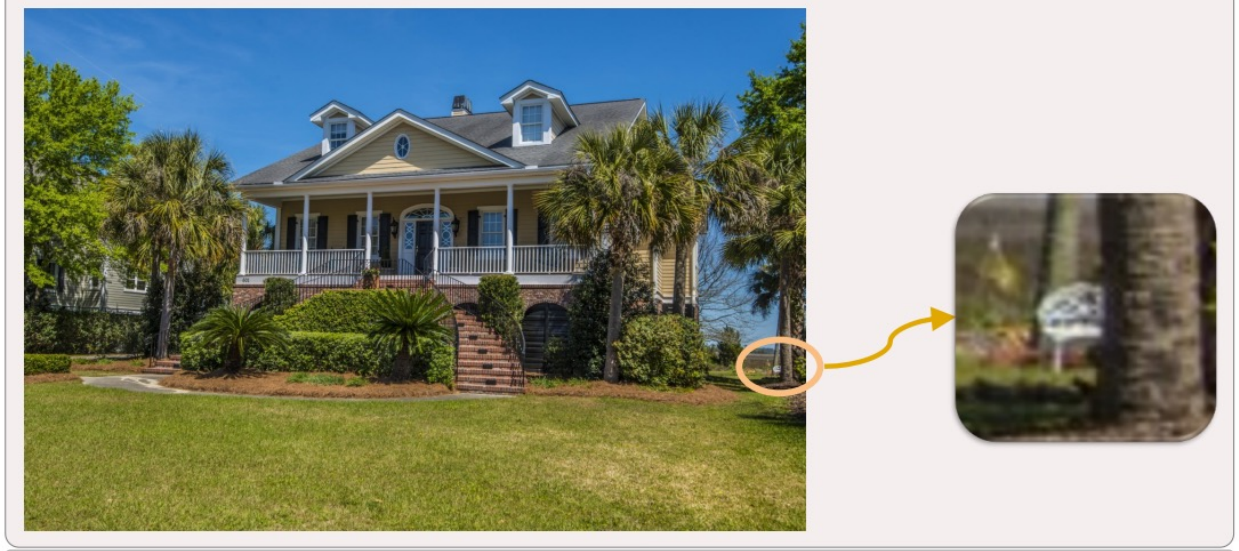
Yifan Z. et al. "Thyme: Think Beyond Images" ICLR 2026.

程序化视觉操作 – Thyme

RL 数据不需要完整的推理过程。

研究团队手工收集了 3 万张高分辨率图像，

挑选出图像中的小物件并构造对应的问题和答案



Question: This picture shows a typical American country-style residence. On the right side of this picture, there is a stool. May I ask what color it is?

Answer: white



Question: What English letters are printed on the bottom of the water cups in the pictures?

Answer: KINTO

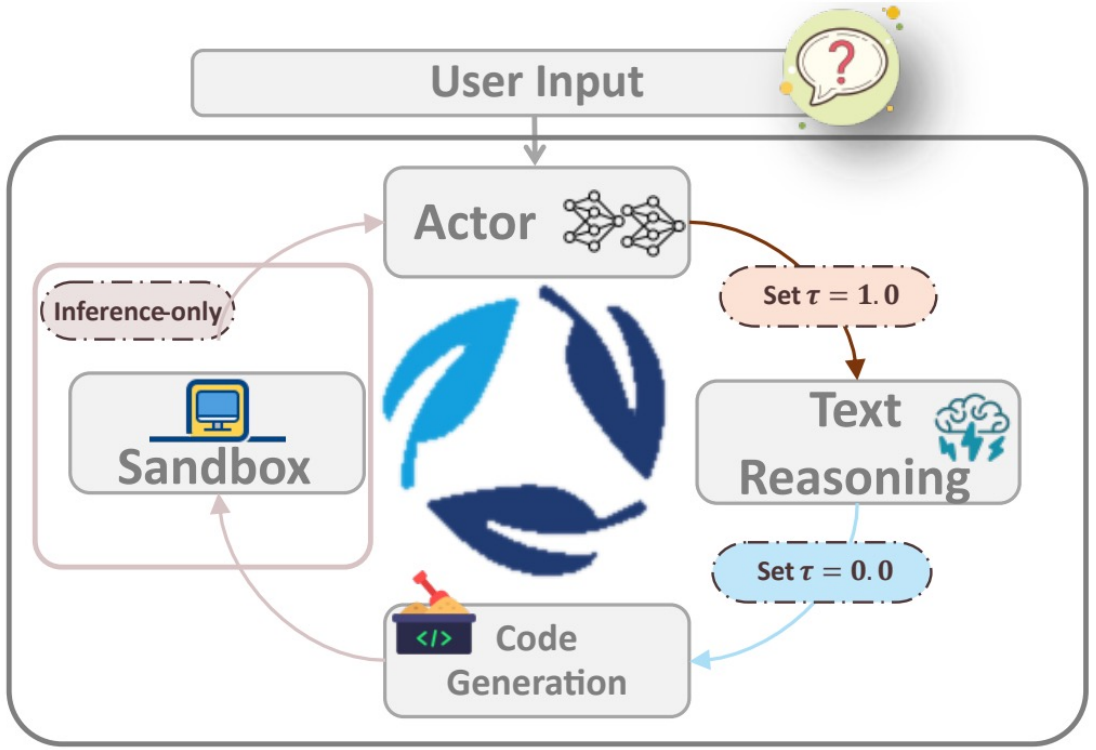
Yifan Z. et al. "Thyme: Think Beyond Images" ICLR 2026.

程序化视觉操作 – Thyme

RL 阶段采用 GRPO 算法，

奖励分为三部分：

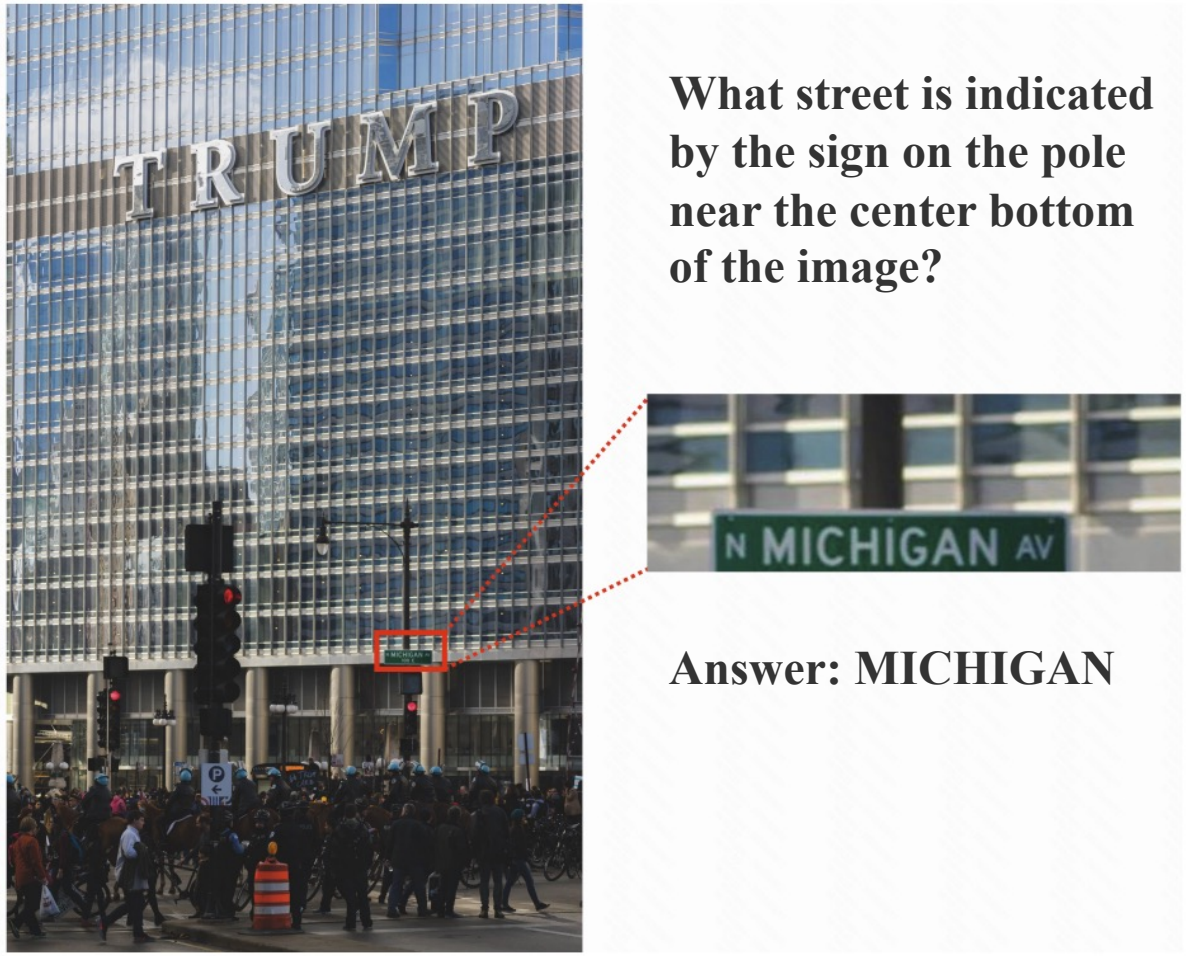
- 格式奖励：回答是否符合格式
- 结果奖励：回答结果是否正确
- 一致性奖励：基于 LLM 验证从过程到答案是否符合逻辑



$$\text{Final Reward } r = \text{Result Reward} \times \left(1 + 0.5 \times \text{Consistency Reward}\right) + 0.5 \times \text{Formatting Reward}.$$

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程序化视觉操作 – Thyme



What street is indicated by the sign on the pole near the center bottom of the image?

Answer: MICHIGAN

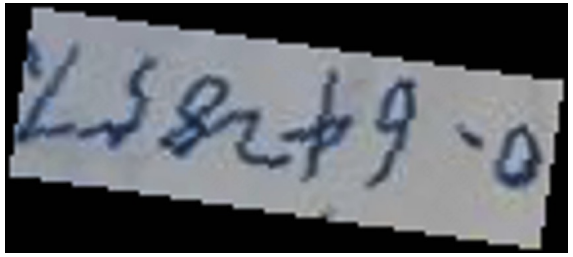
Generated Code

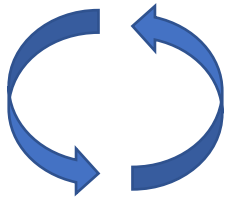
```python
import os
import random
from PIL import Image
# Load the image
image_path = "hr_bench_4k/179.jpg"
image = Image.open(image_path)
# Define the cropping coordinates (x1, y1, x2, y2)
# These coordinates are estimated based on the sign's
# position in the image
x1, y1, x2, y2 = 1550, 2650, 1850, 2750
# Crop the image
cropped_image = image.crop((x1, y1, x2, y2))
# Apply a zoom factor (optional, depending on the required
# level of detail)
zoom_factor = 2
zoomed_image = cropped_image.resize(
    (cropped_image.width * zoom_factor, cropped_image.height * zoom_factor)
)
# Save the processed image with a random suffix
random_suffix = random.randint(1000, 9999)
processed_filename = f"179_{random_suffix}.jpg"
processed_path = "/mnt/data/temp_processed_images/" + processed_filename
zoomed_image.save(processed_path)
# Print the path of the saved cropped image
print(processed_path)
```

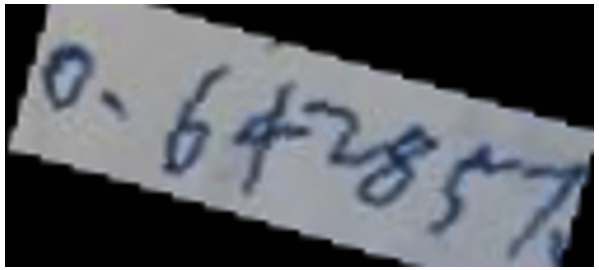
Yifan Z. et al. "Thyme: Think Beyond Images" ICLR 2026.

程序化视觉操作 – Thyme

Please write out the expression of the formula in the image.







Answer: 0.642857

Generated Code

```python
import cv2
import numpy as np
import os
# Load the image
image_path = "OCR_55251.png"
image = cv2.imread(image_path)
# Define the rotation angle (in degrees)
angle = 172 # The angle is derived from the filename suffix
# Get the image dimensions
height, width = image.shape[:2]
# Calculate the rotation matrix
rotation_matrix = cv2.getRotationMatrix2D((width / 2,
height / 2), angle, 1)
# Perform the rotation
rotated_image = cv2.warpAffine(image, rotation_matrix,
(width, height))
# Save the rotated image with a random suffix
random_suffix = np.random.randint(100000, 999999)
processed_path = \
    f"/mnt/data/temp_processed_images/OCR_55251_{random_suffix}.png"
cv2.imwrite(processed_path, rotated_image)
# Print the saved image path
print(processed_path)
```

Yifan Z. et al. "Thyme: Think Beyond Images" ICLR 2026.

程序化视觉操作 – Thyme

7B 的 Thyme 模型在多个评测基准上超过了相近规模的其他模型和 GPT-4o
总体上和 32B 的更大参数模型表现相当

Benchmark	Split	Thyme-VL *7B*	Qwen2.5-VL *7B*	InternVL3 *8B*	Qwen2.5-VL *32B*	GPT-4o
			Perception			
HRbench-4K	FSP	**91.0**+5.8	85.2	78.8	87.5	66.8
	FCP	**63.0**+10.8	52.2	61.3	59.3	63.3
	Overall	**77.0**+8.2	68.8	70.0	73.4	65.0
HRbench-8K	FSP	**86.5**+7.7	78.8	78.8	82.3	60.8
	FCP	57.5+5.7	51.8	**59.8**	58.5	58.5
	Overall	**72.0**+6.7	65.3	69.3	70.4	59.6
MME-Real	Perception	**67.1**+6.5	60.6	63.5	63.8	64.9
	Reasoning	**48.4**+9.8	38.6	44.9	40.4	47.3
	Overall	**64.8**+6.6	58.3	61.3	61.0	62.8
MME-Real-CN	Perception	**70.5**+2.5	68.0	65.5	68.0	63.6
	Reasoning	**52.1**+6.5	45.6	50.0	44.4	51.3
	Overall	**64.6**+3.8	60.8	60.5	60.5	59.7
MME-Real-Lite	Perception	**59.1**+10.4	48.8	51.0	50.6	54.4
	Reasoning	**49.1**+11.3	37.7	44.8	39.3	48.3
	Overall	**55.2**+11.1	44.1	48.6	46.2	52.0
V*	Attribute	**83.5**+5.3	78.2	67.8	77.4	72.2
	Spatial	80.3+6.7	73.6	73.7	**86.8**	60.5
	Overall	**82.2**+5.8	76.4	70.2	81.2	67.5
RealWorld QA	Overall	**70.2**+2.0	68.2	70.0	**70.2**	75.5

Yifan Z. et al. "Thyme: Think Beyond Images" ICLR 2026.

课程提纲

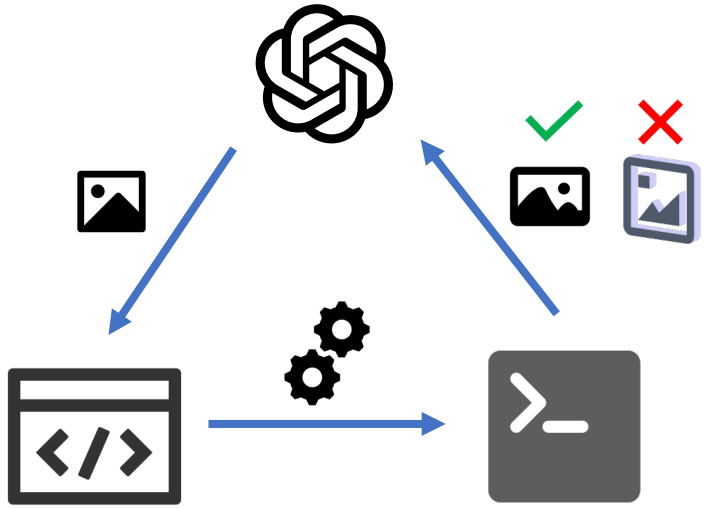
CONTENTS

一、课题背景介绍
二、工具驱动的视觉探索
三、程序化视觉操作
四、原生视觉想象
五、总结与展望

原生视觉想象



Stage (ii): Programmatic Visual Manipulation

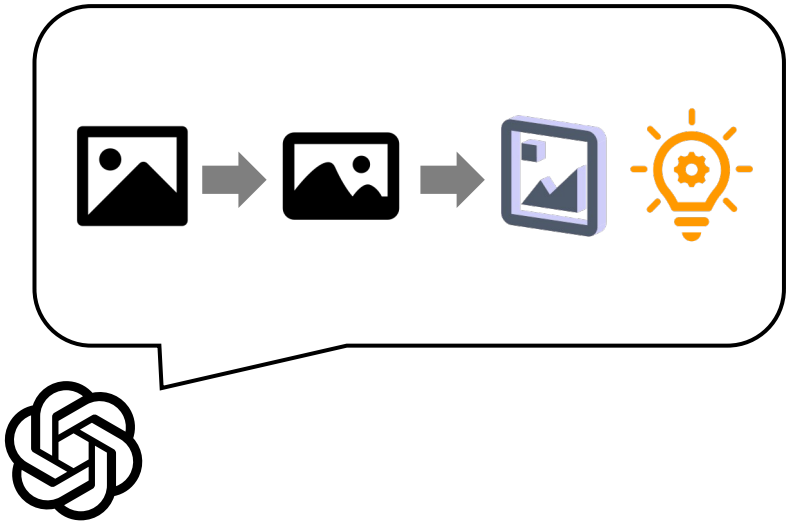


代码对图像的操作

依然存在局限



Stage (iii): Intrinsic Visual Imagination



模型生成中间图像

作为思考过程

原生视觉想象



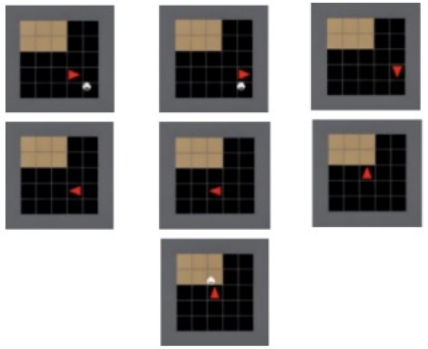
Stage (iii): Intrinsic Visual Imagination

Explicit Visual Reasoning



Q: Can I get the treasure?

Thoughts:

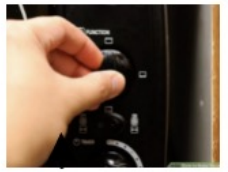


A: Action Success.

Interleaved Multimodal Reasoning



Step 1: Place bread on rack.



Step 2: Turn on oven broiler.

Q: Predict the next steps to make toast.

Thoughts:

[Text] Now, the bread needs to be toasted until golden brown. [Visual]

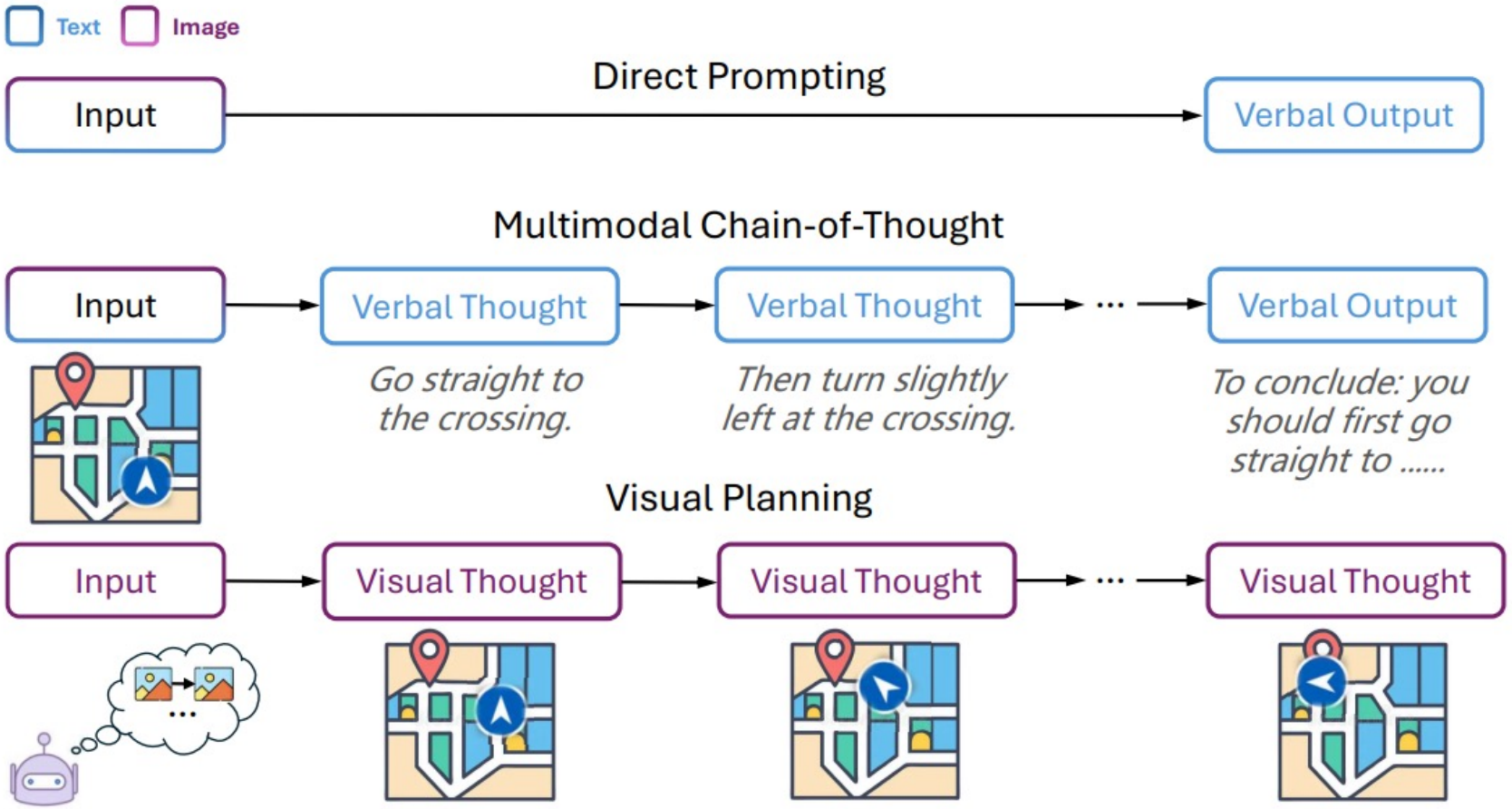


[Text] This shows baking. A better next step is to flip the bread. [Visual]



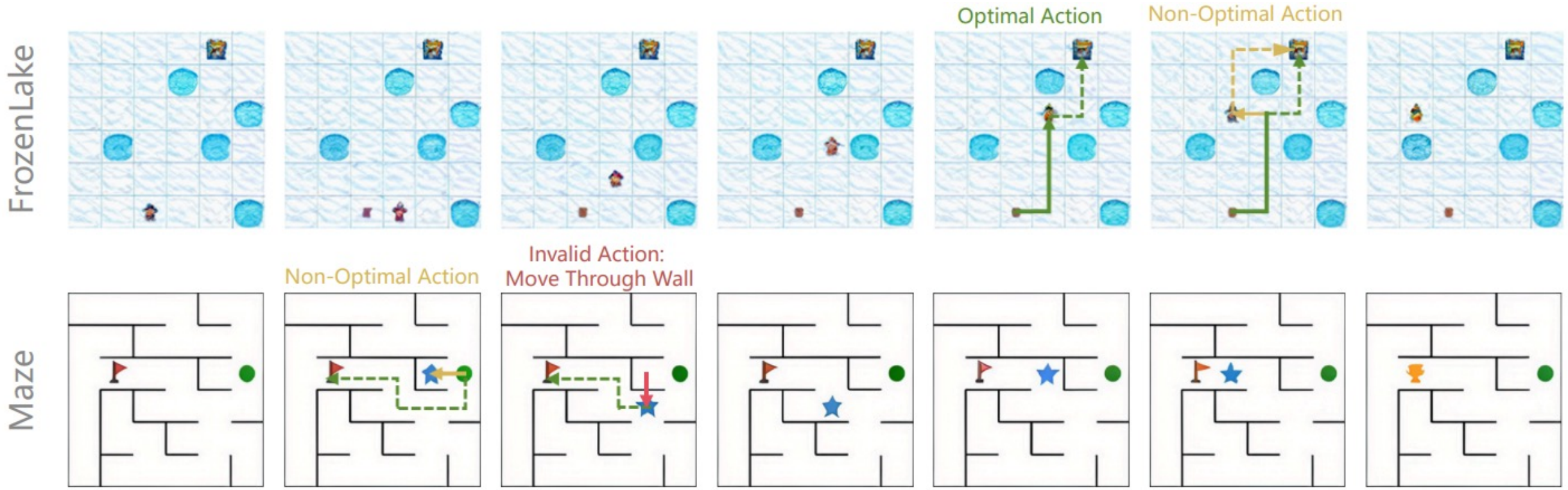
原生视觉想象 – Visual Planning

对依赖视觉信息的任务，图像相比于纯文本可能是更直接有效的推理模态



Yi X. et al. "Visual Planning: Let's Think Only with Images". ICLR 2026.

原生视觉想象 – Visual Planning

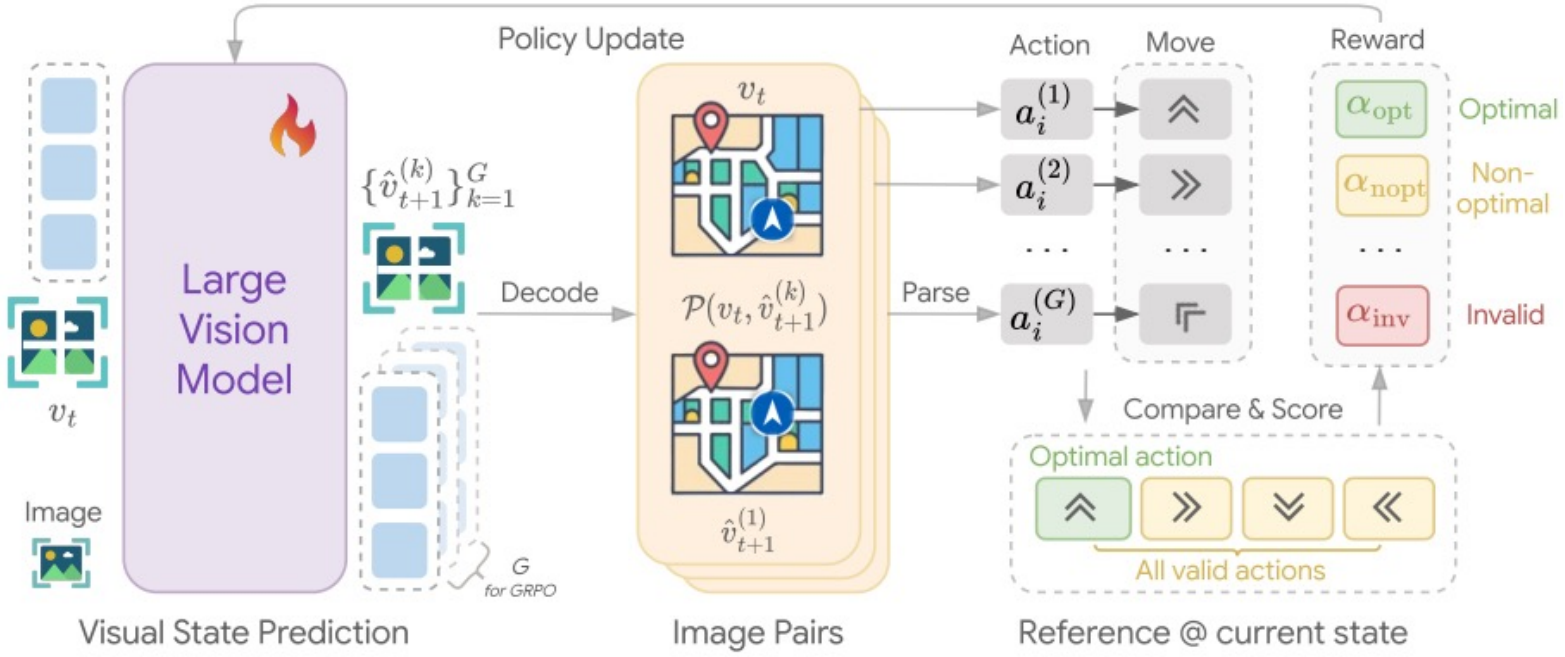


Yi X. et al. "Visual Planning: Let's Think Only with Images". ICLR 2026.

原生视觉想象 – Visual Planning

仅使用图像训练视觉模型：

1. **策略初始化：**用在环境中随机游走的轨迹微调模型，令其具备生成合法图像的能力以及在环境中探索的能力；
2. **强化学习：**通过 GRPO 算法让模型在探索中学会寻找合法的、更优的解法。



Yi X. et al. "Visual Planning: Let's Think Only with Images". ICLR 2026.

原生视觉想象 – Visual Planning

Exact Match (EM) : 模型生成的轨迹是否为最优轨迹

Progress Rate (PR): 模型从起点开始连续正确步骤的比例

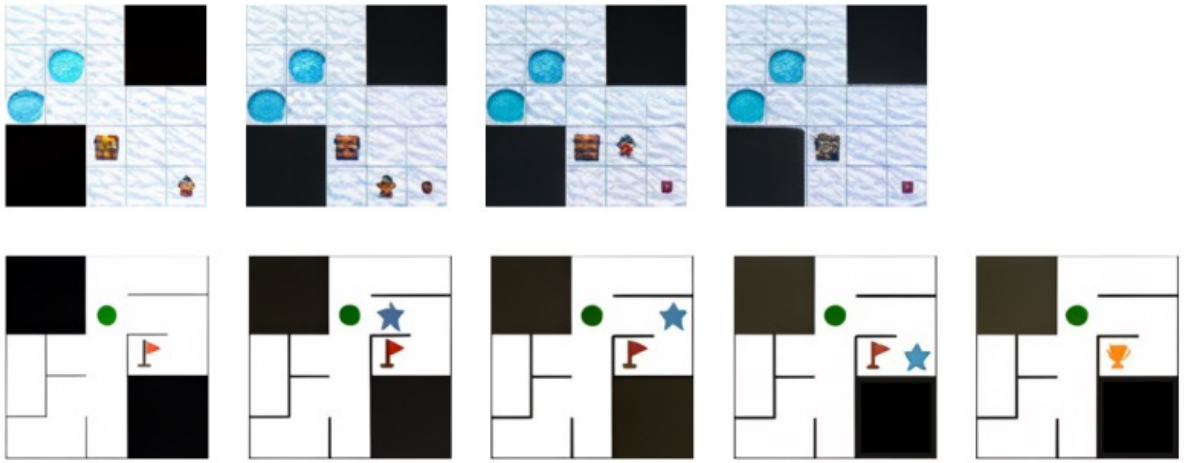
Model	Input	Output	FrozenLake		Maze		MiniBehavior		Avg.	
			EM (%)	PR (%)	EM (%)	PR (%)	EM (%)	PR (%)	EM (%)	PR (%)
Closed-Source Model										
Gemini 2.0 Flash										
- Direct	A+ 🖼	A	21.2	47.6	8.3	31.4	0.7	29.8	10.1	36.3
- CoT	A+ 🖼	A	27.6	52.5	6.9	29.8	4.0	31.2	12.8	37.8
Gemini 2.5 Pro (*think*)	A+ 🖼	A	72.0	85.0	21.5	35.5	37.6	59.9	43.7	60.1
Open-Source Model										
Qwen 2.5-VL-Instruct-7B										
- Direct	A+ 🖼	A	1.2	15.0	0.6	14.5	0.3	9.8	0.7	13.1
- CoT	A+ 🖼	A	8.2	29.1	2.3	15.2	0.5	14.7	3.7	19.7
- SFT†	A+ 🖼	A	68.6	84.4	60.9	70.3	31.3	56.1	53.6	69.9
LVM-7B										
- VPFT† (ours)	🖼	🖼	75.4	79.5	59.0	64.0	33.8	52.2	56.1	65.2
- **VPRL**† (ours)	🖼	🖼	**91.6**	**93.2**	**74.5**	**77.6**	**75.8**	**83.8**	**80.6**	**84.9**

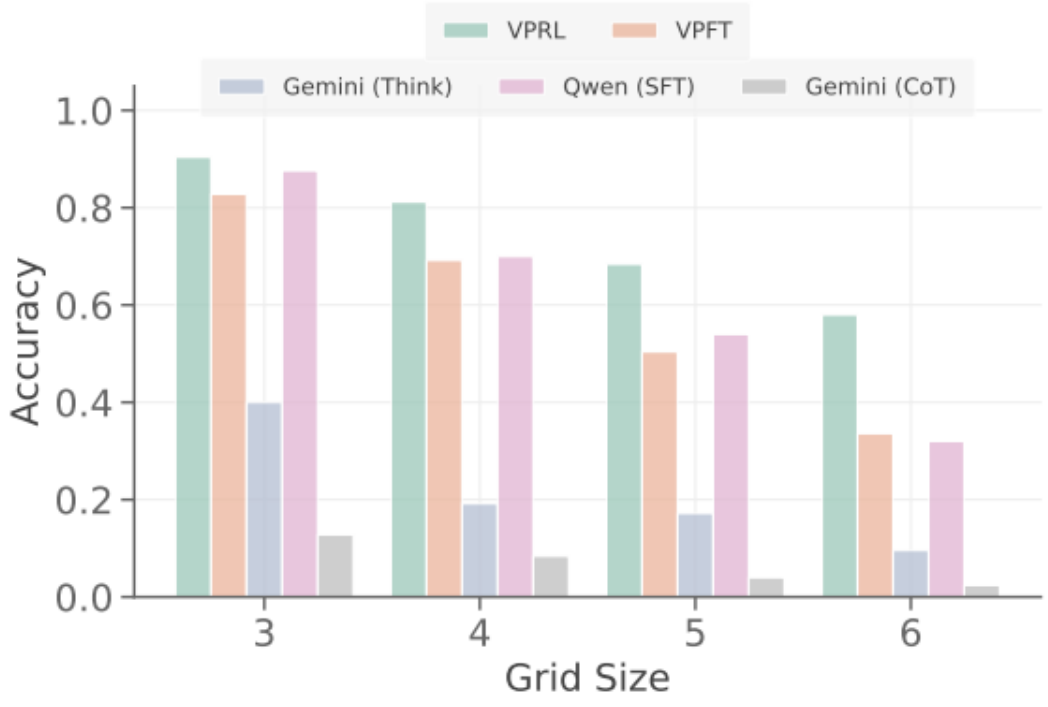
- **SFT:** 对照组，只用文本微调 Qwen2.5-VL 令其输出文本动作序列
- **VPFT:** 对照组，直接使用最优轨迹微调，不进行强化学习
- **VPRL:** 两阶段策略初始化+强化学习

VPRL 效果优于 VPFT 和 SFT, 以及闭源大模型 Gemini

Yi X. et al. "Visual Planning: Let's Think Only with Images". ICLR 2026.

原生视觉想象 – Visual Planning





在输入图片被部分涂黑、网格环境尺寸增大的情况下，

Visual Planner 依然能保持相对稳定的表现

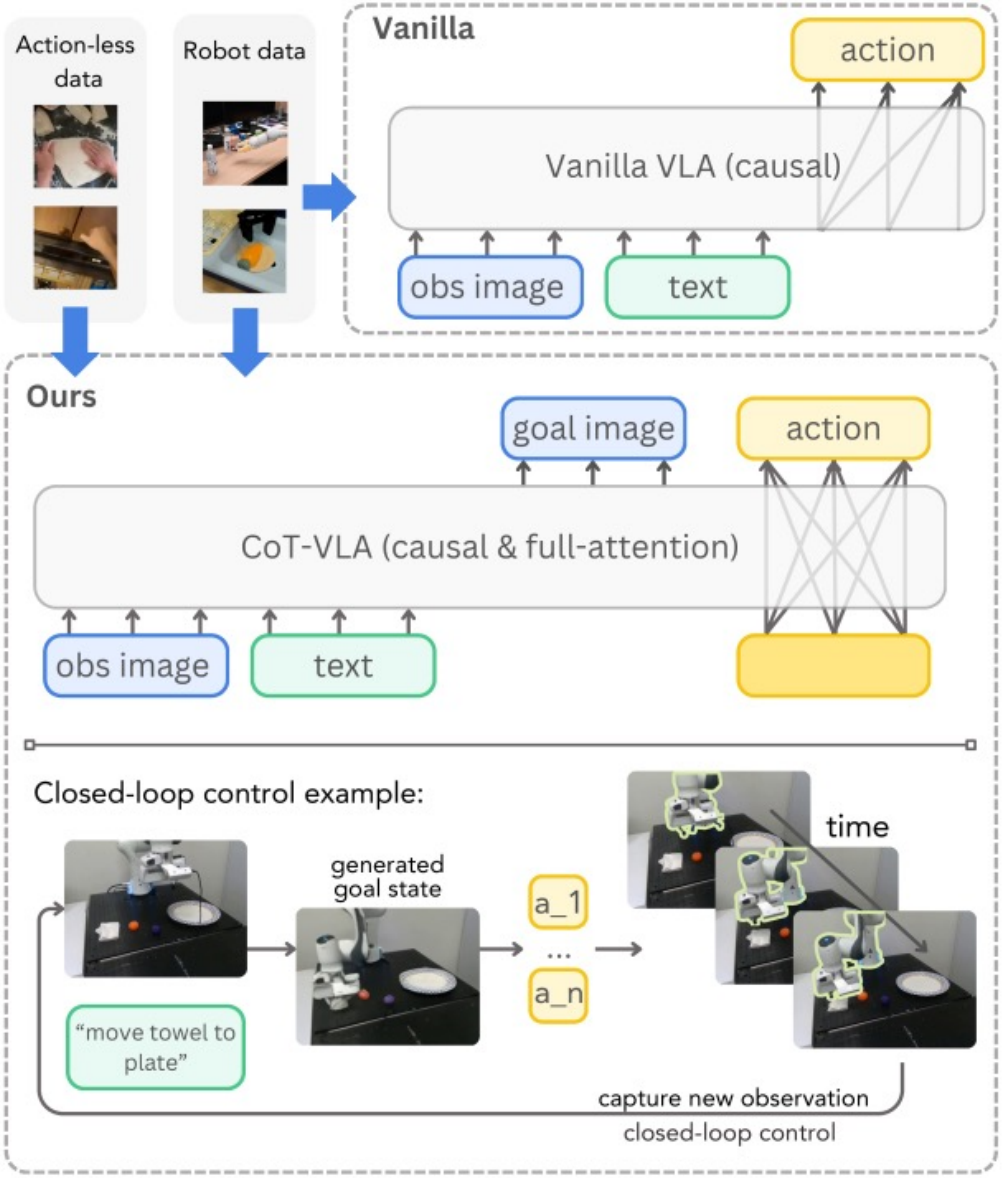
Yi X. et al. "Visual Planning: Let's Think Only with Images". ICLR 2026.

原生视觉想象 – CoT-VLA

Vision-Language-Action (VLA) 模型

- 输入：视觉观测 + 语言指令
- 输出：底层机器人动作

- 传统 VLA：直接生成动作
- CoT-VLA:
  - 预测未来的下一步图像作为子目标
  - 根据当前观测和子目标图像生成动作

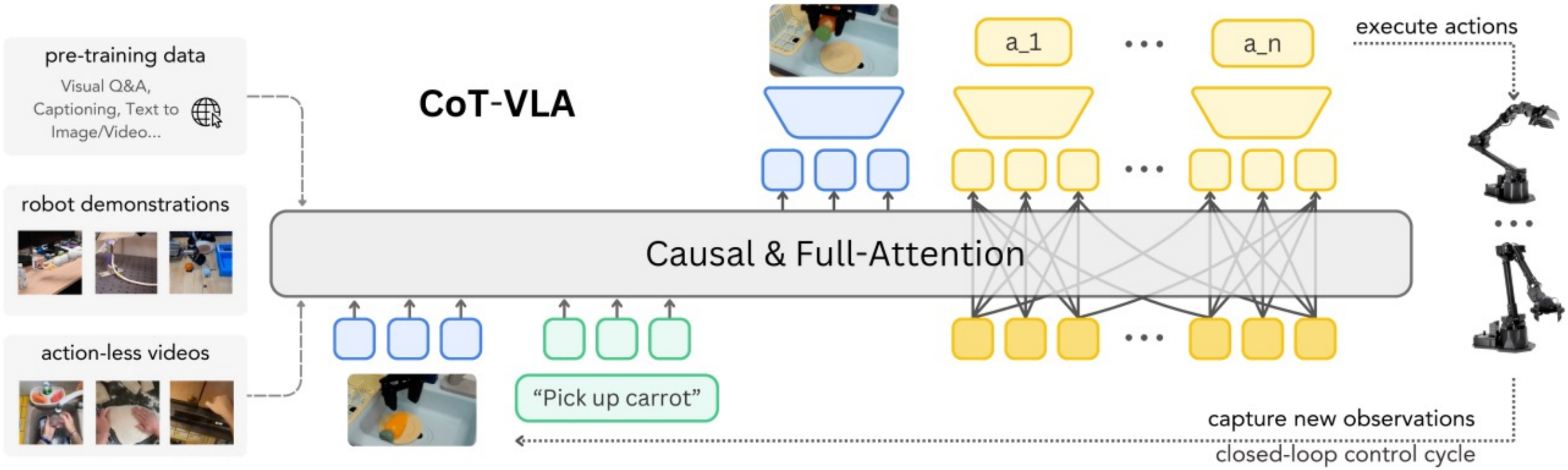


Qingqiing Z. et al. "CoT-VLA: Visual Chain-of-Thought Reasoning for Vision-Language-Action Models". CVPR 2025.

原生视觉想象 – CoT-VLA

以统一视频理解和生成的 VILA-U 模型作为基模，训练模型预测对应的视觉 token 和动作 token

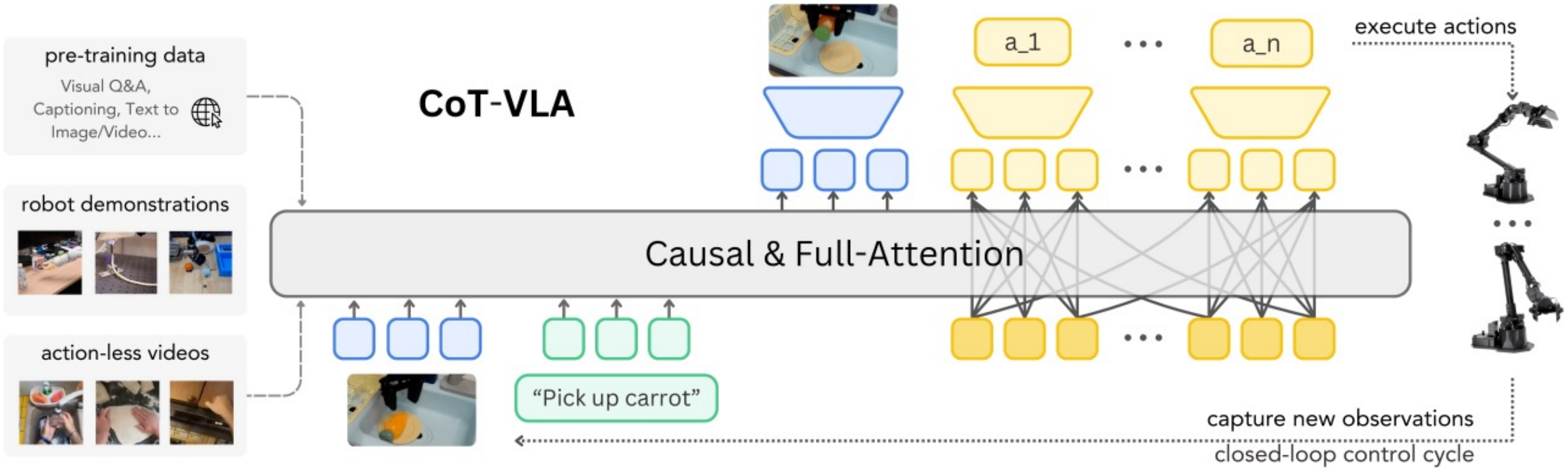
- **传统 VLA 训练需要“语言指令-视觉观测-实机动作”配对的机器人演示数据**
- **CoT-VLA 还可以利用不含机器人动作标注的人类操作视频学习下一步图像的生成**



Qingqiing Z. et al. "CoT-VLA: Visual Chain-of-Thought Reasoning for Vision-Language-Action Models". CVPR 2025.

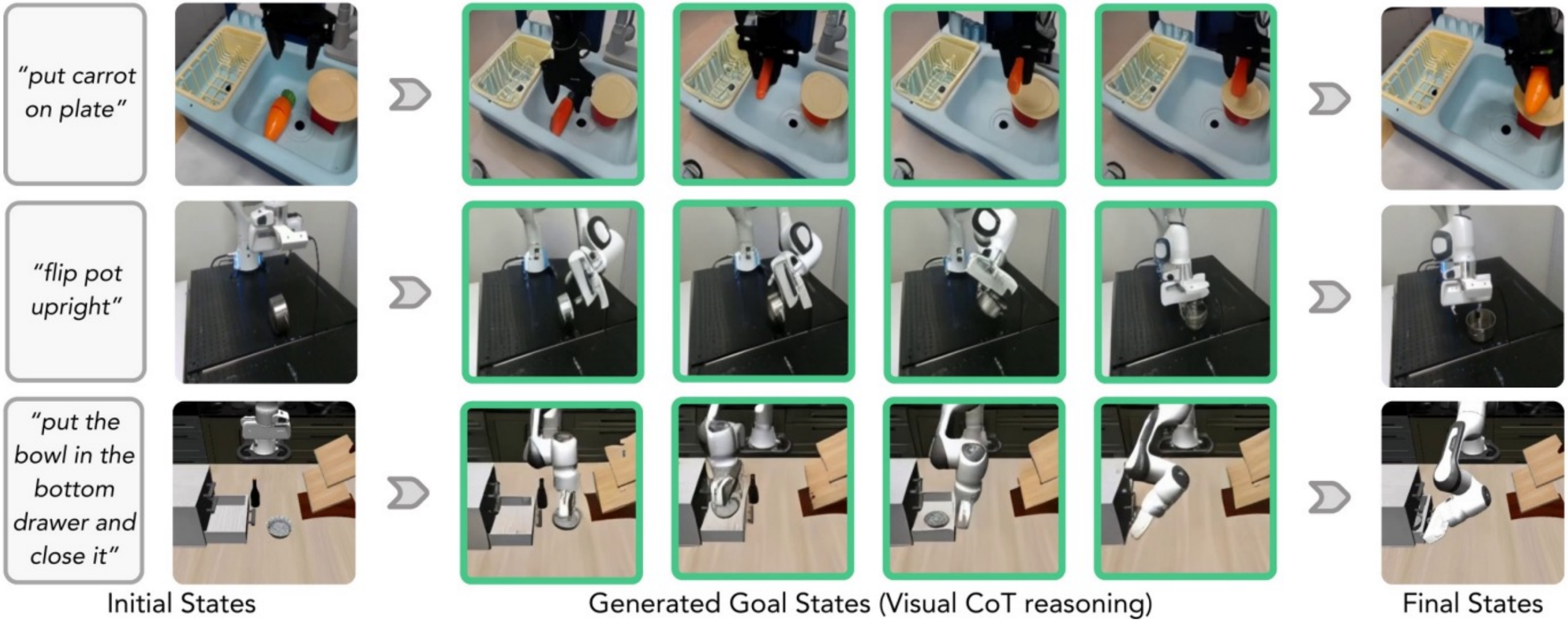
原生视觉想象 – CoT-VLA

- 预训练：
  - 机器人演示数据 Open X-Embodiment
  - 人类操作视频 EPIC-KITCHENS, Something-Something V2
- 下游微调：目标机器人平台的真机数据



Qingqiing Z. et al. "CoT-VLA: Visual Chain-of-Thought Reasoning for Vision-Language-Action Models". CVPR 2025.

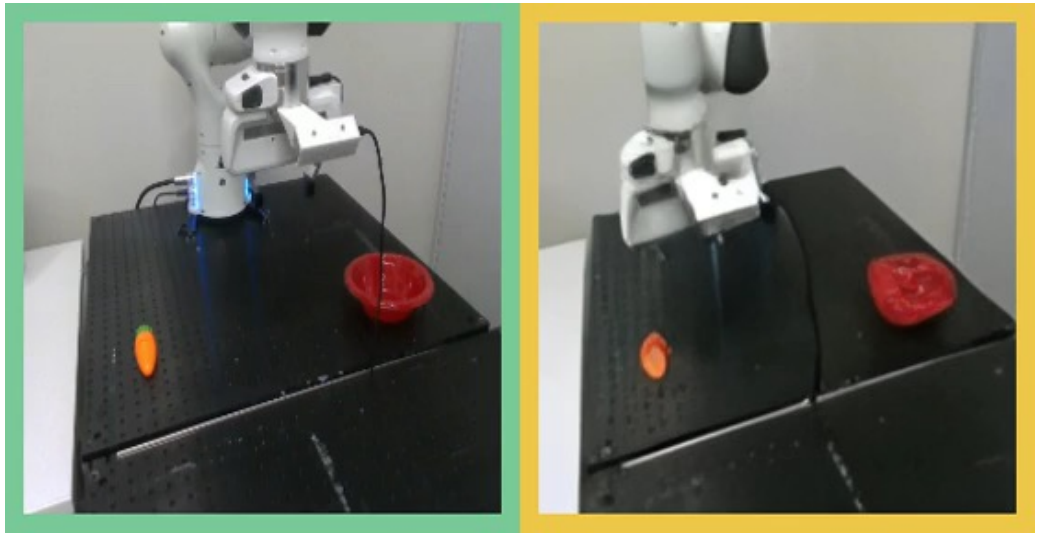
原生视觉想象 – CoT-VLA



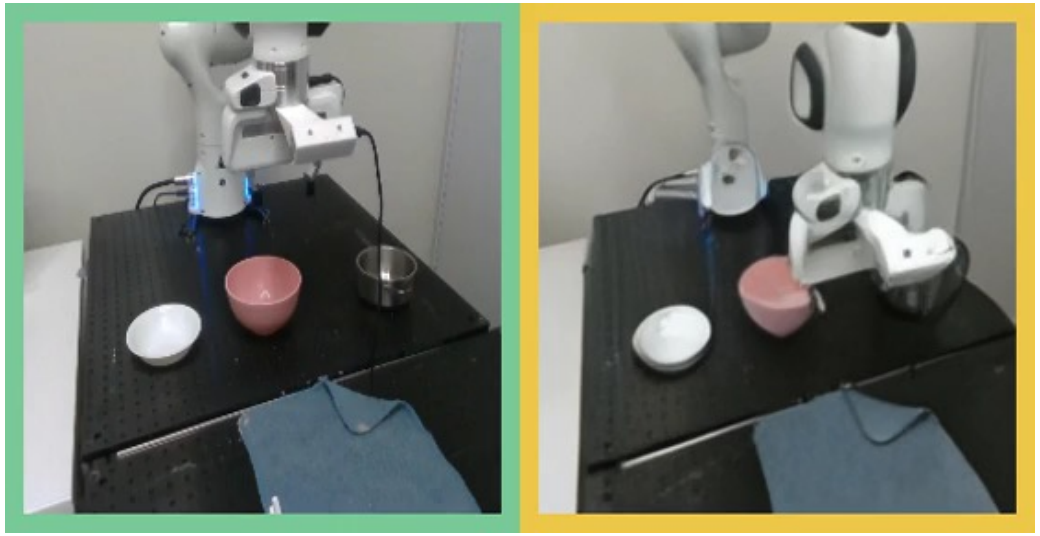
Qingqiing Z. et al. "CoT-VLA: Visual Chain-of-Thought Reasoning for Vision-Language-Action Models". CVPR 2025.

原生视觉想象 – CoT-VLA

绿框为实机视频，黄框为模型生成的系列子目标图像



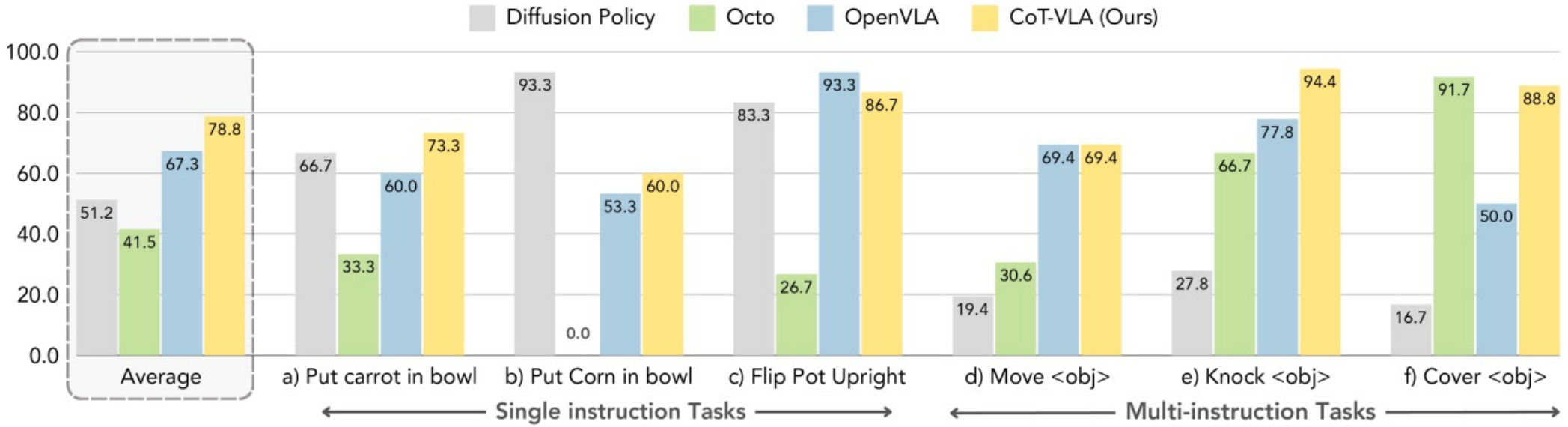
put carrot in bowl

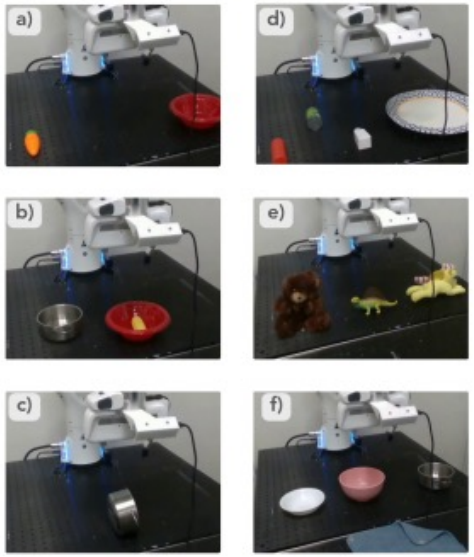


cover white bowl with towel

Qingqiing Z. et al. "CoT-VLA: Visual Chain-of-Thought Reasoning for Vision-Language-Action Models". CVPR 2025.

原生视觉想象 – CoT-VLA





在 Franka-Tabletop 评测基准的六项任务中，

CoT-VLA 取得了最佳的平均性能

Qingqiing Z. et al. "CoT-VLA: Visual Chain-of-Thought Reasoning for Vision-Language-Action Models". CVPR 2025.

原生视觉想象 – FutureSightDrive

FutureSightDrive 提出时空思维链 (Spatio-Temporal CoT)
将视觉想象应用于自动驾驶，通过预测未来帧进行行车轨迹规划



Predicted Future

Spatio-Temporal CoT

Current Observations

Trajectory Planning

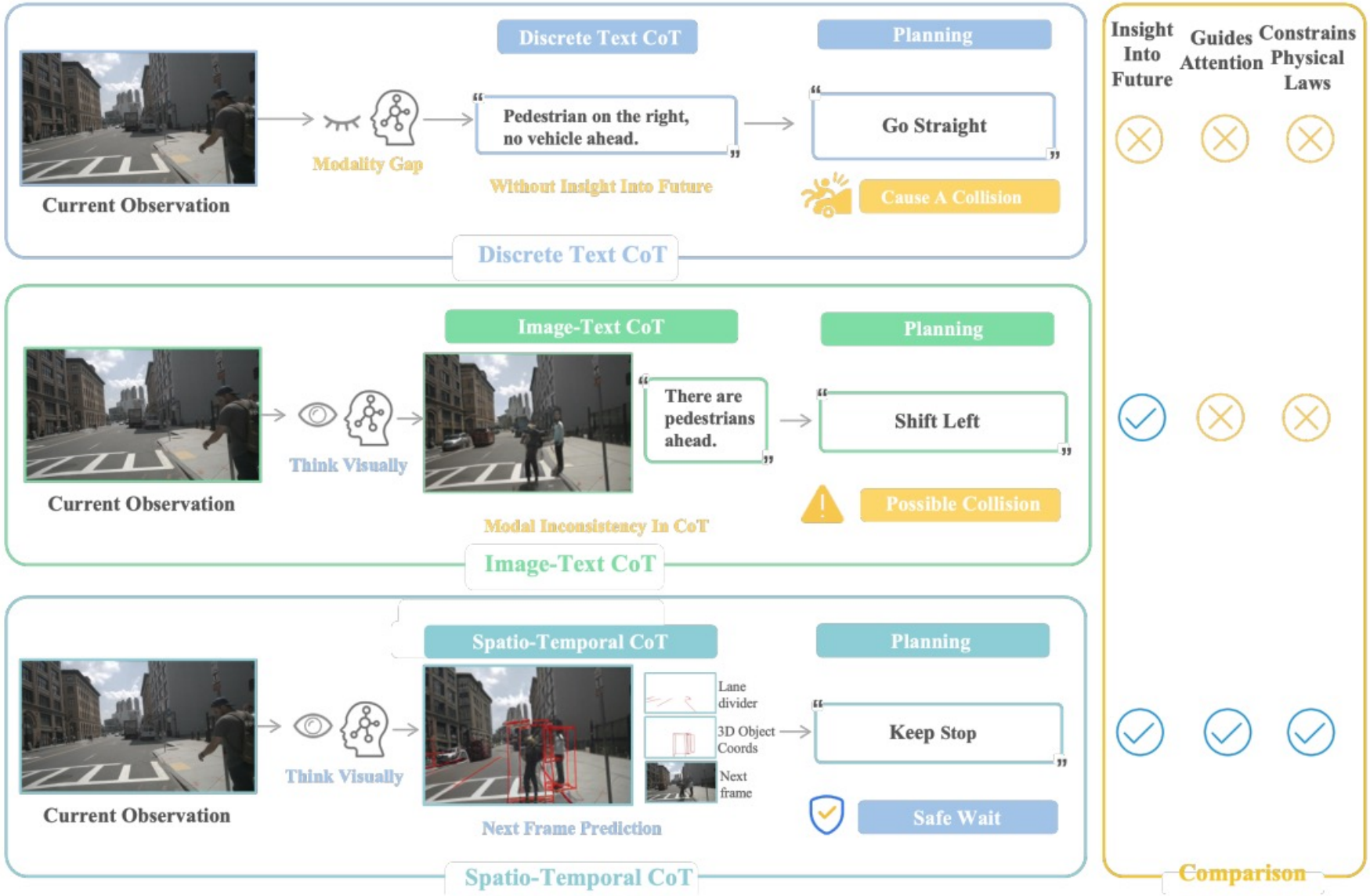
Shuang Z. et al. "FutureSightDrive: Thinking Visually with Spatio-Temporal CoT for Autonomous Driving". NeurIPS 2025.

原生视觉想象 – FutureSightDrive

- **纯文本 CoT**
  - ❌ 文本信息量不足，缺乏时空信息，难以预测未来
- **图文 CoT**
  - ✅ 生成下一帧预测未来
  - ❌ 图文模态缺乏一致性
  - ❌ 缺乏物理约束
- **Spatio-Temporal CoT**
  - ✅ Temporal:

    生成下一帧预测未来
  - ✅ Spatio:

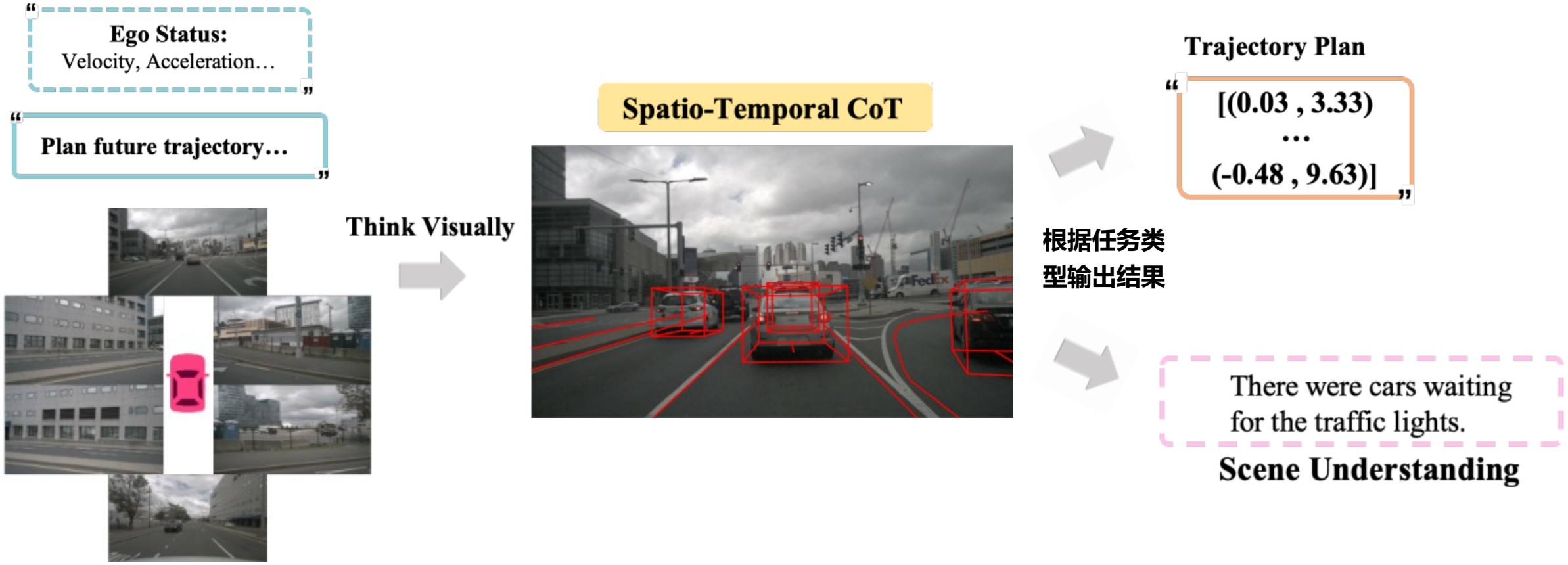
    预测车道线和 3D 物体检测框表示空间关系、引导模型注意力、施加约束



Shuang Z. et al. "FutureSightDrive: Thinking Visually with Spatio-Temporal CoT for Autonomous Driving". NeurIPS 2025.

原生视觉想象 – FutureSightDrive

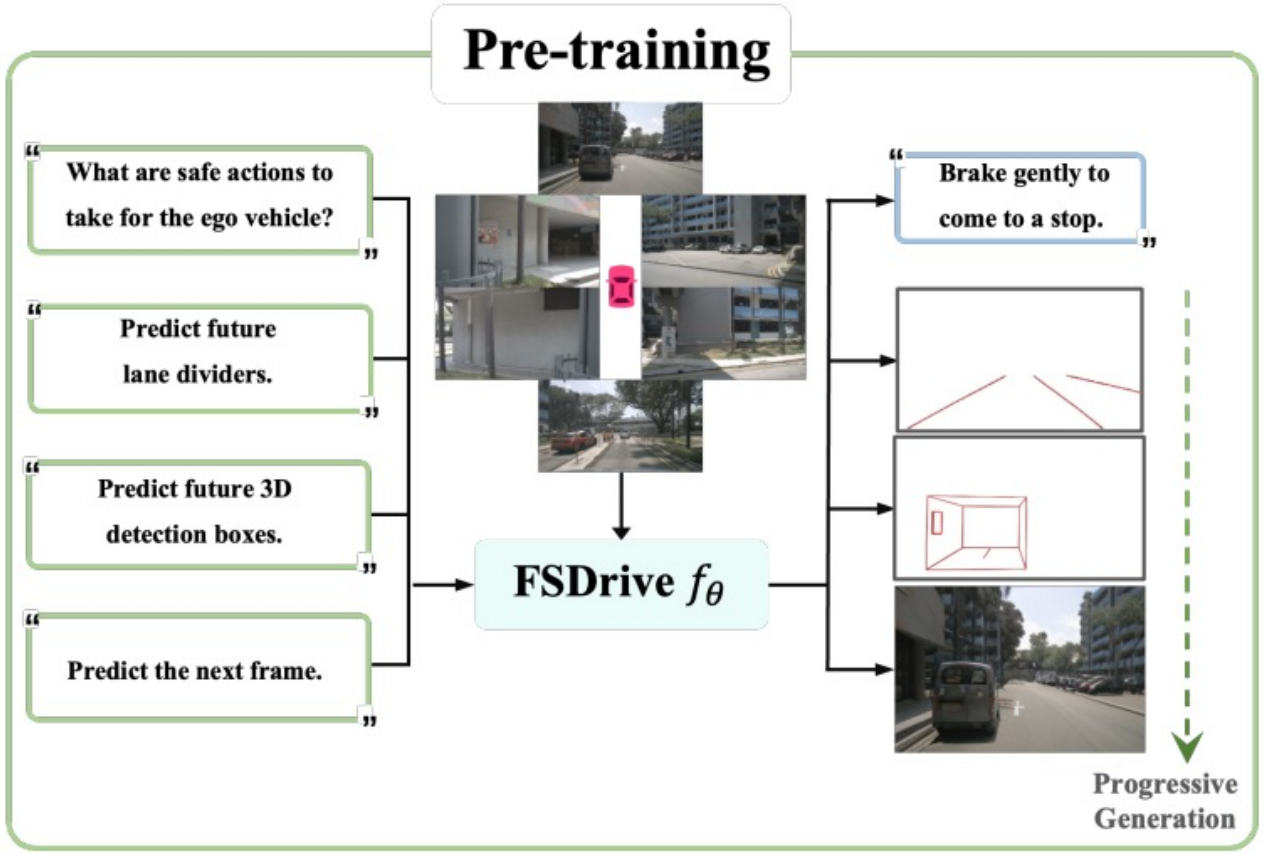
- 输入：汽车四周六个方向摄像头的观测图像 + 任务指令
- 输出：对轨迹规划任务，输出预测轨迹；对场景理解任务，输出场景信息

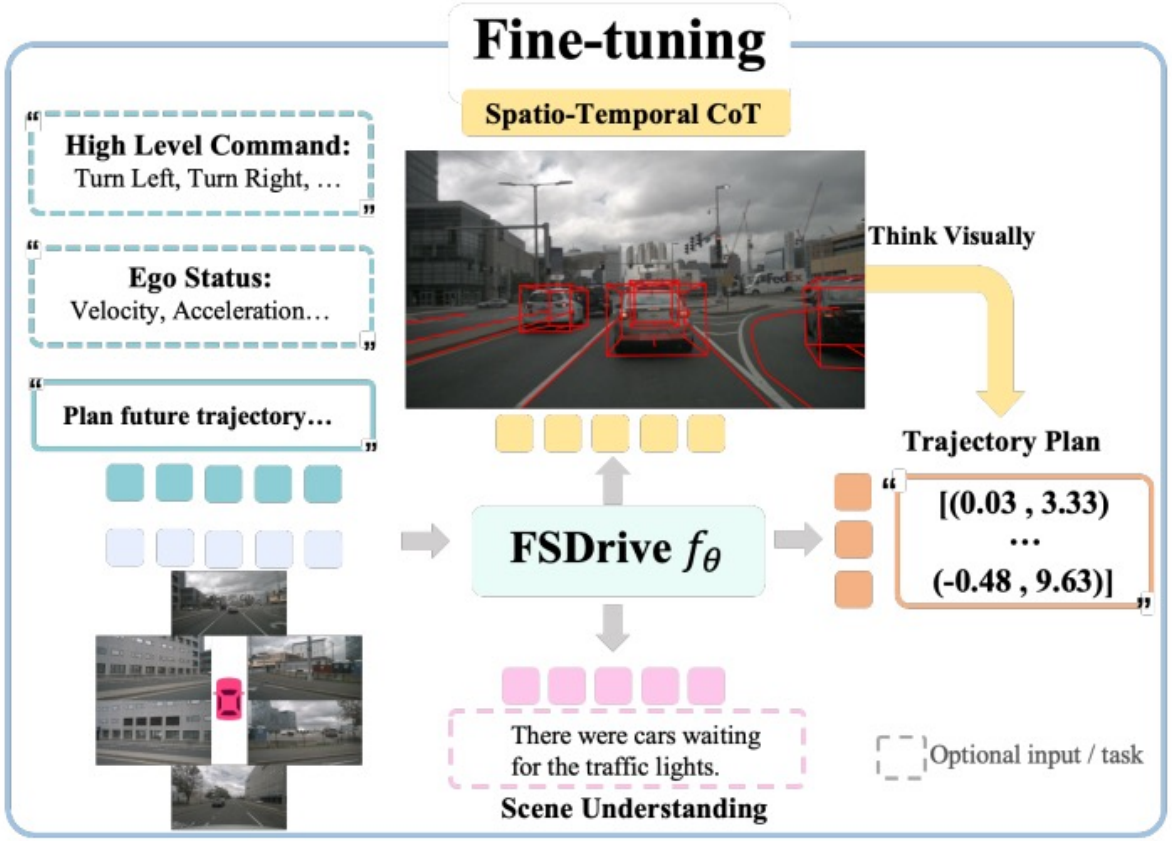


Shuang Z. et al. "FutureSightDrive: Thinking Visually with Spatio-Temporal CoT for Autonomous Driving". NeurIPS 2025.

原生视觉想象 – FutureSightDrive

- 预训练：基于驾驶数据激活模型生成下一帧以及车道线、3D 检测框的能力
- 微调：基于场景理解数据和轨迹规划数据训练模型对应任务的能力





Shuang Z. et al. "FutureSightDrive: Thinking Visually with Spatio-Temporal CoT for Autonomous Driving". NeurIPS 2025.

原生视觉想象 – FutureSightDrive

消融实验表明：

- **预测未来帧、3D 检测框和车道线有助于提高轨迹预测的准确率并减少碰撞概率；**
- **Spatio-Temporal CoT 优于无 CoT 和其他 CoT 方法；**

VQA	Future frames	Future 3D detection	Future lane divider	L2 (m) ↓ 1s	2s	3s	Avg.	Collision (%) ↓ 1s	2s	3s	Avg.
×	×	×	×	0.45	1.09	2.12	1.22	0.12	0.43	1.45	0.67
✓	×	×	×	0.46	1.07	2.04	1.19	0.12	0.42	1.42	0.65
×	✓	×	×	**0.39**	0.96	1.71	1.02	0.10	0.38	**1.32**	0.60
×	×	✓	×	0.46	1.06	1.99	1.17	0.10	0.37	1.35	0.61
×	×	×	✓	0.42	0.97	1.80	1.06	0.13	0.41	1.40	0.65
✓	✓	✓	✓	**0.39**	**0.91**	**1.63**	**0.98**	**0.09**	**0.36**	1.33	**0.58**

Type	L2 (m) ↓ 1s	2s	3s	Avg.	Collision (%) ↓ 1s	2s	3s	Avg.
None	0.39	0.91	1.63	0.98	0.09	0.36	1.33	0.58
Text CoT	0.39	0.92	1.61	0.97	0.10	0.29	1.21	0.53
Image-text CoT	**0.38**	0.90	1.65	0.98	0.09	0.25	1.15	0.50
Spatio-temporal CoT	0.40	**0.89**	**1.60**	**0.96**	**0.07**	**0.12**	**1.02**	**0.40**

Shuang Z. et al. "FutureSightDrive: Thinking Visually with Spatio-Temporal CoT for Autonomous Driving". NeurIPS 2025.

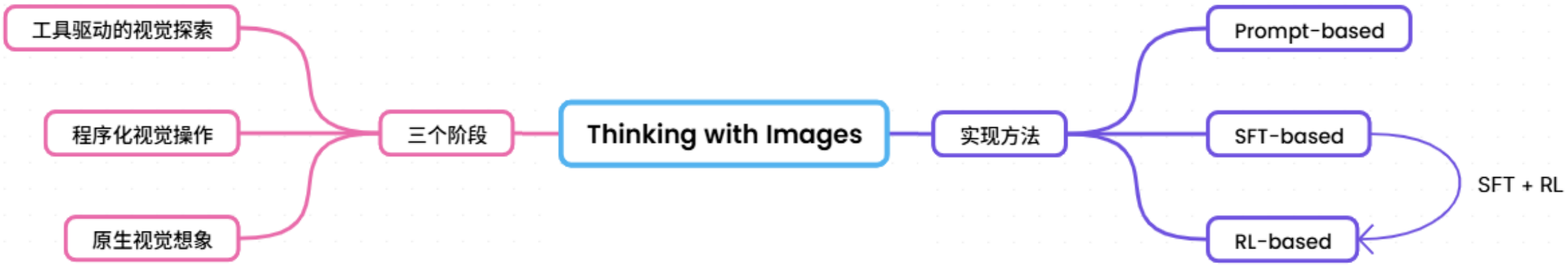
课程提纲

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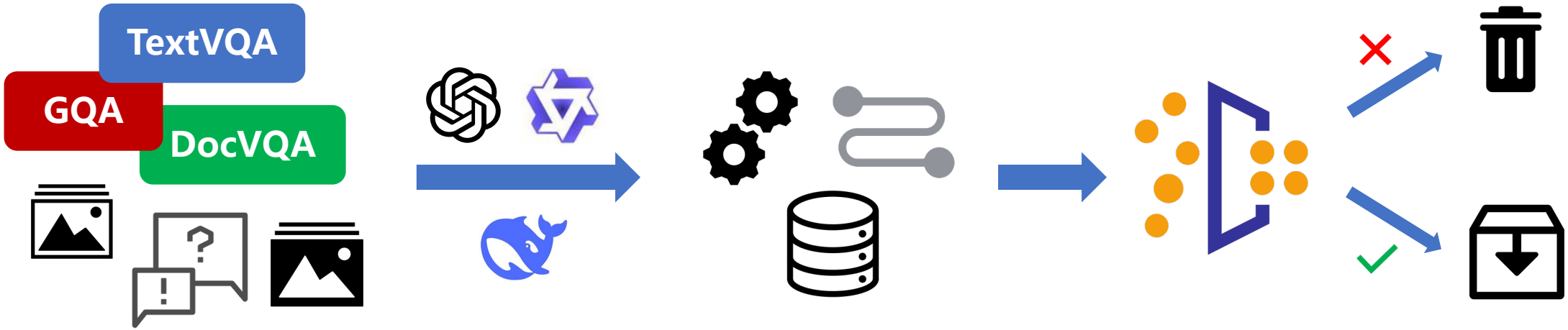
总结与展望

三个阶段 × 三种实现框架



工具驱动的视觉探索 – VLM-R³

借助大模型生成稀缺的完整推理链数据



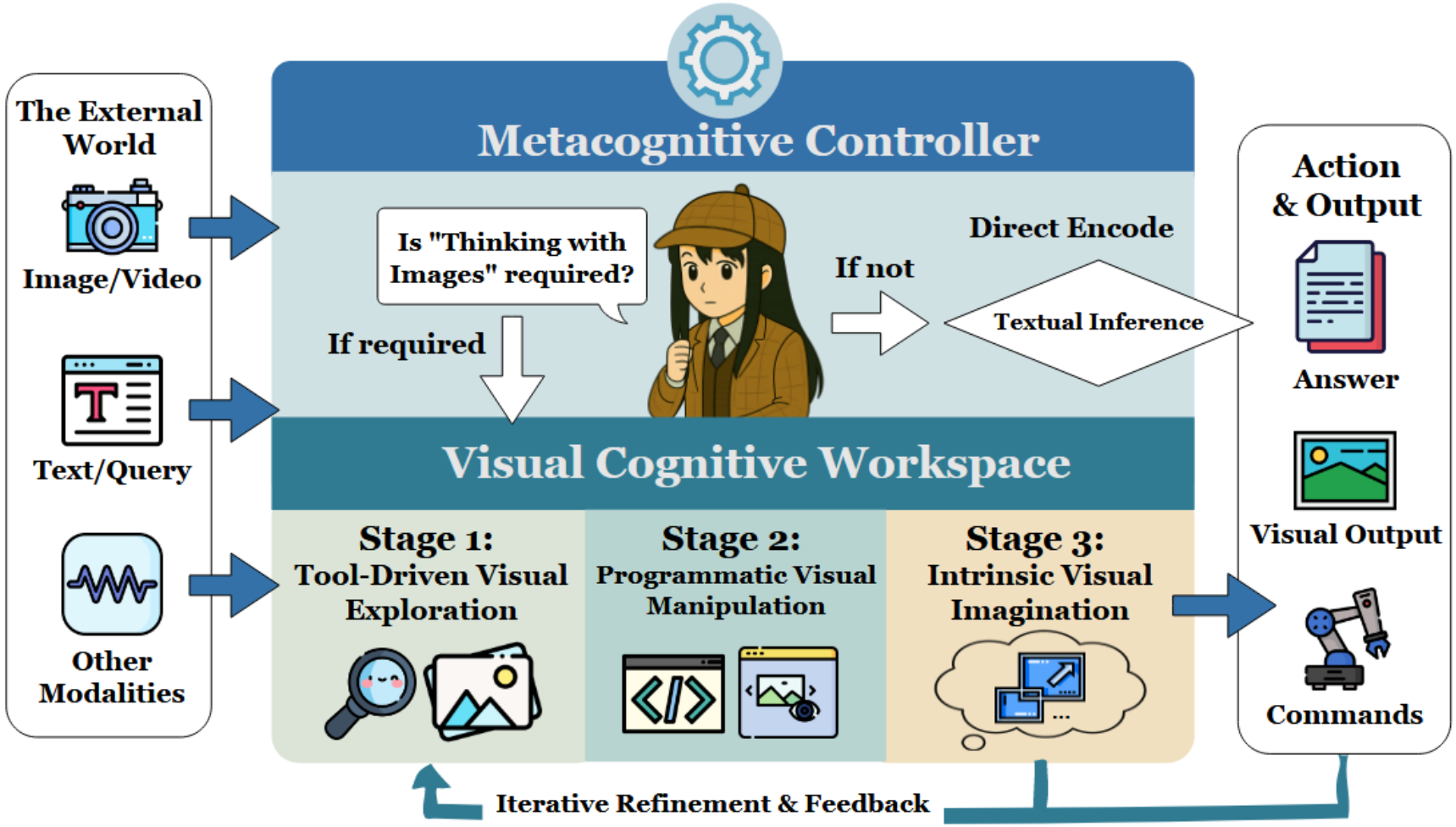
公开数据作为“种子”

借助大参数模型
生成符合期望格式的推理链

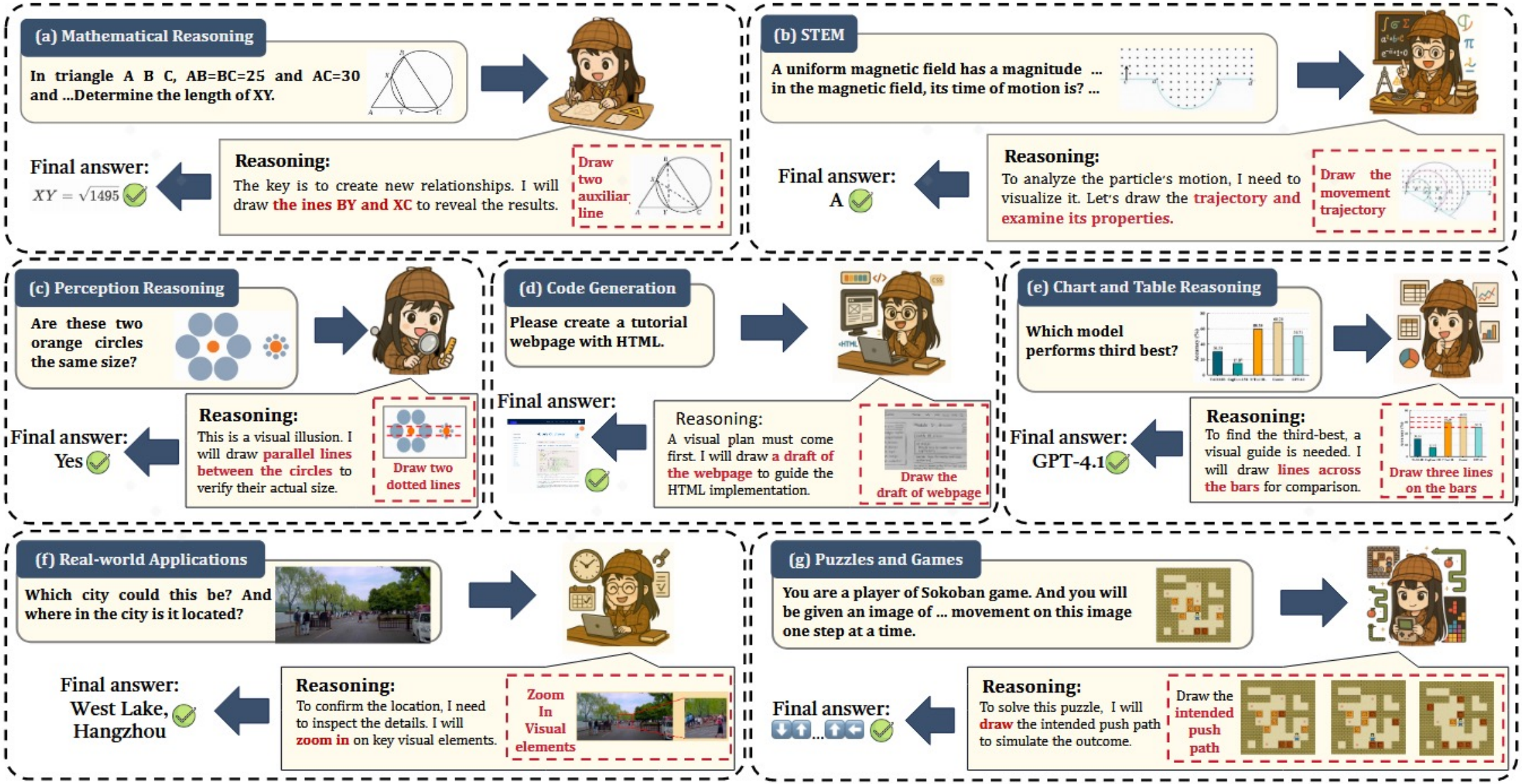
人工/规则/大模型进行验证
过滤低质量数据

总结与展望

Thinking with Images 的统一框架

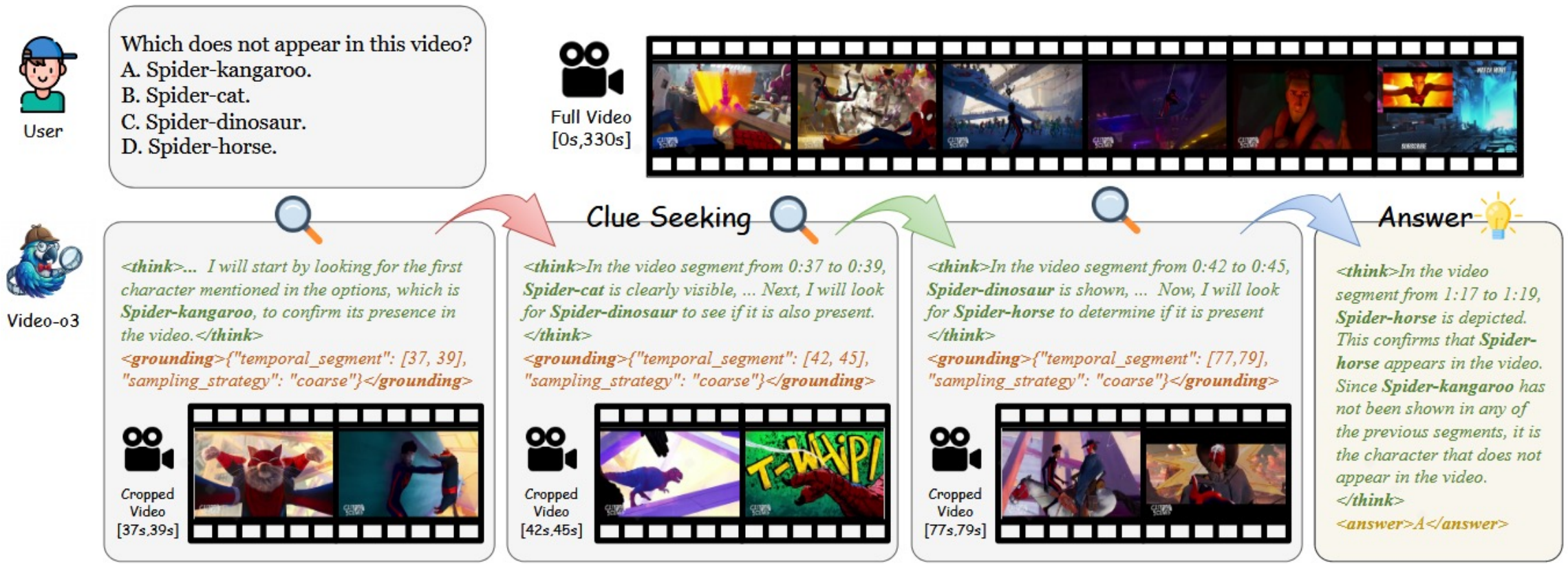


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(a) Mathematical Reasoning
In triangle A B C, AB=BC=25 and AC=30 and ...Determine the length of XY.
Final answer:
$XY = \sqrt{1495}$
Reasoning:
The key is to create new relationships. I will draw the ines BY and XC to reveal the results.
Draw two auxiliar line
(b) STEM
A uniform magnetic field has a magnitude ... in the magnetic field, its time of motion is? ...
Final answer:
A
Reasoning:
To analyze the particle's motion, I need to visualize it. Let's draw the trajectory and examine its properties.
Draw the movement trajectory
(c) Perception Reasoning
Are these two orange circles the same size?
Final answer:
Yes
Reasoning:
This is a visual illusion. I will draw parallel lines between the circles to verify their actual size.
Draw two dotted lines
(d) Code Generation
Please create a tutorial webpage with HTML.
Final answer:
Reasoning:
A visual plan must come first. I will draw a draft of the webpage to guide the HTML implementation.
Draw the draft of webpage
(e) Chart and Table Reasoning
Which model performs third best?
Final answer:
GPT-4.1
Reasoning:
To find the third-best, a visual guide is needed. I will draw lines across the bars for comparison.
Draw three lines on the bars
(f) Real-world Applications
Which city could this be? And where in the city is it located?
Final answer:
West Lake, Hangzhou
Reasoning:
To confirm the location, I need to inspect the details. I will zoom in on key visual elements.
Zoom In Visual elements
(g) Puzzles and Games
You are a player of Sokoban game. And you will be given an image of ... movement on this image one step at a time.
Final answer:
Reasoning:
To solve this puzzle, I will draw the intended push path to simulate the outcome.
Draw the intended push path

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从 Thinking with Images 到 Thinking with Videos



Xiangyu Z. et al. "Video-o3: Native Interleaved Clue Seeking for Long Video Multi-Hop Reasoning". arxiv:2601.23224

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其他未来方向：

- **更高效的视觉思考**
  - **按需自适应思考，根据难度动态调节思考成本……**
- **视觉推理安全**
  - **防推理链投毒、工具投毒、误导思考……**
- **可以验证中间推理过程的新评测基准**
  - **不仅验证最终答案，还评估思维链的质量**
- **Thinking with Audio/Video, Thinking with World…**
  - **从图像扩展到音频、视频、与环境的具身交互……**
- **……**

课程提纲

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一、课题背景介绍
二、工具驱动的视觉探索
三、程序化视觉操作
四、原生视觉想象
五、总结与展望

感谢聆听！



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